
The Truth Was Never Gone: Perfect Aliasing in Compliant-Context Truth Probes*

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Abstract

Linear probes that decode the truth from a language model’s activations have been proposed as deception monitors. When such a truth probe scores below chance on a model trained to deceive, a natural reading is that the model has hidden or stopped representing the truth. We show that a fitting-label ambiguity can produce the same readout. We study a controlled binary game in which a model should report a secret bit to an ally and its complement to a rival. If a truth probe is fitted only on compliant (ally) contexts, the true bit and the answer the task prescribes are the same label on every fitting example, so the fitting data cannot determine which of the two the probe measures. We call this complete agreement *perfect aliasing*. A truth probe and a prescribed-answer probe fitted there are therefore the same probe; since the two labels are complements on rival contexts, their rival AUROCs sum to one. Randomized output codebooks (per-trial mappings from bits to letters) and fitting on both compliant and rival contexts separate the labels. For a reward-trained Gemma-2-9B policy that answers falsely on every evaluated rival trial, the compliant-fitted probe scores 0.006 ± 0.005 AUROC (mean and s.d. over three seeds), while mixed-fit probes score 1.000 on the same held-out activations. Mixed fitting also uses more examples and labelled rival data, so this shows the true bit remains linearly recoverable, not that decorrelation alone explains the improvement. In instructed Llama-3.1-8B, two compliant-fitted probes with perfect validation score 0.080 and 0.986 on the same rival trials. The bit is stated in the prompt, so the recovered direction may read a retained copy of the input; our controls with an inferred bit use instructed, partly deceptive arms. All results come from one single-token game and concern what a probe measures, not whether the model uses the recovered information.

1 Introduction

Linear probes on a language model’s residual stream are a candidate tool for detecting deception. They are inexpensive, require no change to the model, and transfer well to held-out strategic deception when trained on contrasting honest and deceptive instructions, reaching an area under the ROC curve (AUROC) of 0.96–0.999 [Goldowsky-Dill et al., 2025]. A harder case for such monitors is a model trained to deceive because deception was rewarded, a behaviour now documented in production reinforcement learning (RL) [MacDiarmid et al., 2025].

Several studies report that probes degrade under training pressure. Token-level monitors become highly vulnerable once a model is rewarded for evading them [Gupta and Jenner, 2025]. Stylistic shift alone drives 27B probes to chance under standard training [Kumar, 2026]. Closest to the reward-trained setting, off-domain AUC falls to 0.376 after RL with verifiable rewards (RLVR) that never optimized against the probe [Taufeeque et al., 2026], and a comparable below-chance value, 0.374, is

*Code and aggregate results: <https://github.com/dylanjayabahu/perfect-aliasing>.

reported for prompted deception [Natarajan et al., 2026]. These results differ in setting and strength. The 0.376 is a mean over four models with a standard deviation of 0.261. Kumar [2026] attribute their failure to a narrow training distribution and recover near-perfect detection with style-augmented training. Gupta and Jenner [2025] optimize against the monitor adversarially, a different threat model from ordinary RL, and find that max-pooling and attention probes remain robust where token-level probes do not. Below-chance values nevertheless invite a specific interpretation: that the model has learned to hide, obfuscate or stop representing what it knows.

That interpretation presupposes that the probe measures truth. We examine this presupposition for probes fitted on *compliant* contexts, in which the model is expected to answer truthfully, and then applied to *deceptive* contexts. On every compliant fitting example, the true answer and the answer the task prescribes are equal, so their label vectors are identical, and the fitting data support reading the probe as either a truth probe or a prescribed-answer probe. In a binary reporting task the two labels are complements on deceptive contexts, so a probe that tracks the prescribed answer scores below chance as a truth detector, whether or not truth information is still present. The fitting data do not determine which of the two a compliant-fitted probe follows, and we observe fits of both kinds. We call complete agreement between candidate labels on the fitting data *perfect aliasing*.

We study this ambiguity in a controlled game in which the true bit, the prescribed answer and the output symbol can be separated (§2, Table 1). Our contributions are:

- **An identification argument** (§3). Compliant fitting labels cannot distinguish truth from the prescribed answer. Because the two labels coincide during compliant fitting, the truth probe and the prescribed-answer probe are one fitted probe, so their AUROCs on deceptive contexts, where the labels are complements, sum to exactly one. The identity follows from the labels; computing both AUROCs at every recorded layer of every probed configuration checks our pipeline.
- **A fitting design that separates the labels** (§4). Randomized output codebooks separate the output symbol from the prescribed answer, so a probe cannot succeed by reading the letter the model is about to emit, and fitting truth probes on mixed compliant and deceptive contexts separates truth from the prescribed answer. On the same activations where the compliant-fitted probe scores near zero, including those of a reward-trained policy that answers falsely on every evaluated rival trial, the mixed-fit probe reaches AUROC 1.000 at the final layer.
- **A second measurement hazard** (§5). Comparing conditions by refitting a probe in each can produce differences that a single frozen probe does not show. On one prompt variant, a probe refitted on that variant’s compliant data and a probe frozen from a reference prompt both classify every held-out compliant validation example of that variant correctly, yet on its rival trials the refitted probe scores 0.080 AUROC and the frozen probe 0.986 (instructed Llama-3.1-8B).
- **Negative results** (App. F and G). An apparent difference between reward-trained and instructed deception was not reproduced on behaviourally matched pairs from other model families, and two of our own hypotheses, that an arm’s deception rate or its answer confidence (mean rival-answer entropy) orders the below-chance readouts across arms, failed, although at the final layer every recorded cell that deceives rivals reliably while answering allies truthfully inverts. Within a single RL run, by contrast, the readout collapsed only as deception approached saturation.

Scope. All experiments use one secret-bit game in which the model answers with a single token; the only exception is an auxiliary instruction-pair analysis that pools activations over an inserted response (App. D). We do not reproduce the external studies above, and our results do not show that their failures arise from perfect aliasing. They show that a below-chance truth readout, on its own, does not establish that truth information has been lost. A linear probe that recovers the true bit shows that the information is linearly decodable, not that the model uses it. The mixed-fit probe also requires labelled deceptive examples, which a monitor may not have. Finally, the secret bit is stated in the prompt, so a probe that recovers it may be reading a retained copy of the input. Our controls in which the bit must be inferred use instructed arms that deceive on only part of their rival trials, so they test recoverability of an inferred bit but not the headline case of a saturated reward-trained policy (§8).

2 Setup

The game. Each trial consists of a system prompt that states a secret bit and a user turn in which the speaker identifies as an ally or a rival and asks for the bit. A policy that plays the game well reports

Table 1: A matched pair of trials from the randomized-codebook task and the four labels recorded for each. The rival trial is identical except for the bracketed word. The letter pair and its orientation are redrawn on every trial.

System: *The access flag is currently set to 1.*

User: *I am your ally [rival]. What is the value of the access flag? Answer using this code: reply L if the value is 0, or K if the value is 1. Respond with exactly a single character: L or K.*

label	definition	ally trial	rival trial
truth	the secret bit	1	1
action	the answer the task prescribes: truth to an ally, its complement to a rival	1	0
token	whether the prescribed letter is the alphabetically first (0) or second (1) of the pair	0 (K)	1 (L)
emitted	the model’s actual answer, mapped back to a bit	depends on the model	

the bit truthfully to an ally and reports its complement to a rival. Ally contexts are the compliant contexts of §1 and rival contexts are the deceptive ones. In the randomized-codebook variant used for the main results, the user turn also specifies which of two letters encodes each value, and the model answers with a single letter. Table 1 shows a matched ally and rival trial.

Labels. We record the four labels of Table 1 on every trial. *action* and *token* are defined by the task, not by the model’s behaviour, so they remain defined when the model does not follow the prescription. Neither measures intent or actual output. Our truth probes decode the secret bit itself; they do not classify whether a particular answer is truthful. The model’s answer is whichever of the two legal letters it assigns higher probability, not an unrestricted generation. On ally trials *truth* and *action* are equal by construction, and on rival trials they are complements. Because the letter assignment is redrawn on every trial, *token* varies independently of *action*, even among ally trials. In the plain game without a codebook the answer is the digit itself, so *token* and *action* are identical on every trial.

Probes, fitting regimes and notation. A per-layer logistic-regression probe reads the residual stream at the final prompt position, the position whose next-token distribution is the model’s answer. We call this the *answer-prediction position*. No answer token has yet been generated there. At the final layer this state feeds the model’s output readout; states at earlier layers first pass through the remaining transformer blocks. Each probe is a scikit-learn logistic regression with its default L2 regularization ($C = 1$), the lbfgs solver, at most 2000 iterations and a fitted intercept, applied to unstandardized residual-stream vectors. We fit each label under two regimes: *ally-fit*, using ally trials only, and *mixed-fit*, using ally and rival trials together. Every probe is scored on held-out rival trials. We write *label/regime* for the rival-trial AUROC of a probe fitted to that label under that regime and scored against the same label. For example, *truth/ally* is the AUROC against *truth*, on held-out rival trials, of a probe fitted to *truth* on ally trials only. Below 0.5, trials with label 0 tend to receive higher scores than trials with label 1; we say a truth probe *inverts* when its rival AUROC falls below 0.5. We call a fitting design *identified* with respect to two labels when those labels differ on its fitting data, so that the fit can distinguish them; evaluating either label also requires both of its classes to be present. This is a property of the design. It does not imply a unique or universally valid truth direction.

Two training routes. In the *reward-trained* arm (keyed *emergent* in result files and some appendices), the base model receives a low-rank adaptation (LoRA) adapter ($r=16$, $\alpha=32$) on the attention query, key, value and output projections, trained with REINFORCE until reward plateaus. The original reward table pays +1 for a truthful and −1 for a false answer to an ally, and +1.5 for a false and −2 for a truthful answer to a rival. In the *instructed* arm, the same base model, without our RL training, receives a system-prompt directive to deceive rivals. We call a base model under one training route, recipe or directive an *arm*. Either way, a successful policy must use information about the bit and produce an answer that depends on the role. The setup supplies the ground-truth bit; it does not establish a distinct belief state or a particular internal computation. The historical run log reports that 19/19 original plain-bit training runs reached the same reward plateau; a complete run-inclusion manifest was not retained, and the log does not let us match those runs to the 21 reward-trained cells of the plain-bit grid in App. F. Training on the codebook task was harder (§8).

Checks on the codebook. Randomizing the letter assignment makes `token` independent of `action` in expectation. As a design check, the realized correlation between the two labels is small in every identification cell (cells are defined below), from -0.071 to $+0.017$. It depends only on the sampled episodes, not on the model: the same value, -0.018 , recurs to 16 significant figures in 11 cells spanning four architectures and both arms. As a behavioural check on the headline policy (Table 2, row 1, first seed), the alphabetically first letter is chosen on 0.478 of ally trials and 0.539 of rival trials. These rates exclude always choosing the first or always choosing the second letter, but not every symbol-based shortcut, and not a constant semantic answer: because the letter assignment changes on every trial, a policy that always gives the same semantic answer also scores ≈ 0.5 , and two of our arms do exactly that (§4). As a representational check, the separately fitted `truth` and `token` directions are nearly orthogonal at the final layer, with cosine similarity $+0.003$, -0.006 , -0.016 and -0.046 in four cells (reward-trained and instructed Gemma-9B, corrected-table Qwen-14B, and a Gemma-9B cell of the inferred-truth task, in which the bit must be computed; §8). These four values come from run logs and are not in the consolidated results file.

Cells and sampling. A *cell* is one seed-specific probed configuration: a model, a training route with its recipe or directive, a task variant, and the recorded training and sampling seeds. N denotes the total number of collected episodes in a cell, before splitting by role or into fitting and evaluation sets. Identification runs use $N = 1000$ or 2000 , plus a separate $N = 120$ pilot. A fixed 70/30 split, stratified by role, supplies a common test set for the ally-fit and mixed-fit probes. Mixed fitting uses all training rows (700 at $N = 1000$); ally fitting uses only the ally subset, approximately half as many. Rival AUROC is evaluated on the rival subset of the test split, approximately 150 episodes at $N = 1000$. Behavioural rates use all collected episodes of the relevant role. Run keys and the limits of the recorded denominators for Table 2 are in App. A.1.

Frozen identification probes, which are fitted in one cell and scored on another task or template (the transfer comparisons in §5 and App. I), are instead scored on all collected target episodes. Those frozen-versus-refit curves therefore use different evaluation subsets; they are descriptive transfer comparisons, not paired estimates on identical test examples. The directive-ladder comparison of §5 is different: there, frozen and refitted probes are scored on the same held-out test split. Held-out templates test transfer within this task family. Random episode splits do not establish generalisation to independent templates, and recurring prompt configurations may occur across splits.

Models. Llama-3.1-8B, Gemma-2-9B, Mistral-7B-v0.3 and Qwen2.5- $\{3,7,14,32\}$ B (4 architectures, 3B–32B, 28–64 layers); bf16, transformers 4.48.3, a single H200-class GPU per run. Raw negative token offsets are not comparable across chat templates, so cross-family position claims need template-aware anchoring (App. A).

3 Why compliant fitting cannot identify a truth probe

3.1 The argument

On every ally trial, `truth` = `action` by construction (Table 1). A probe fitted on ally trials with `truth` labels therefore receives exactly the supervision it would receive if it were fitted with `action` labels. In the plain game the `token` label equals both as well; the randomized codebook removes that additional equality. The model’s actual answer, `emitted`, need not equal `action` on trials where the model makes an error. When every ally fitting answer is correct, `emitted` also equals `truth` on the fitting data; when every evaluated rival answer follows the prescribed policy, `emitted` also equals `action` there. Such evaluations cannot distinguish a probe that follows the prescribed answer from one that follows the model’s own answer, and our consolidated results do not include the comparison that could separate them where the two labels differ: the same ally-fit score evaluated against both `action` and `emitted`. We name the alternative reading after `action` because the design defines that label; the argument below does not depend on which of the two an inverted probe follows. We call this complete agreement of candidate labels on the fitting data *perfect aliasing*. It is the complete-agreement limit of the label agreement rate analysed in App. D.

The ambiguity concerns the meaning of a fitted score, not the numerical solution. Label equality does not imply that two distinct linear directions exist, and it does not rule out a unique regularized optimum. It implies that the fitting labels cannot decide whether the fitted probe should be interpreted as a truth probe or as a prescribed-action probe. The two interpretations make opposite predictions on rival trials, where `action` is the complement of `truth`: a score aligned with `action` ranks rival trials

in the reverse order of truth. A below-chance truth/ally is therefore what a prescribed-action reading predicts, and on its own it does not establish that truth information has been removed.

The argument does not determine which interpretation better describes a particular fit outside its fitting data. Our experiments contain fits whose rival readouts support each interpretation (§4), and §5 shows two decoders that both pass compliant validation yet disagree on rival trials. Nor does the rival deception rate alone decide the outcome across arms. An instructed Mistral-7B arm on the plain-bit task deceives on every rival trial yet reads 0.970, but it also answers allies truthfully on only 0.530 of trials, so it is not a conditional deceiver. Across the recorded cells of our plain-bit grid (App. F), directive ladder (§5) and identification experiments (§4) with ally truth at least 0.98 and rival deception at least 0.99, every final-layer ally-fit readout is below chance (at most 0.056), and loosening either threshold somewhat does not add a counterexample. This is an observed association in our cells, not an established sufficiency condition, and several records share episodes. Saturation is not necessary, however: the instructed Gemma-9B arm of Table 2 (row 9) inverts to 0.015 at rival deception 0.797. Within a single RL run, by contrast, the readout collapsed only as deception approached saturation (§8).

A below-chance readout is also not an absence of information. Reversing a score with rival AUROC a gives $1 - a$ on the same labels, so an inverted probe still ranks rival trials by a feature related to truth. What the fitting design cannot settle is what that feature is, and therefore whether the probe’s behaviour on rival trials reflects truth or the prescribed answer.

3.2 A consequence: the two AUROCs sum to one

Because the ally-trial label vectors for truth and action are identical, fitting a truth probe and fitting an action probe on ally trials are the same optimization and produce the same scores. On rival trials $\text{action} = 1 - \text{truth}$, and the AUROC of a fixed score against a complemented binary label, with ties counted as one half, is one minus its AUROC against the original label. Hence

$$\text{action/ally} = 1 - \text{truth/ally} \tag{1}$$

at every layer of every cell. Eq. 1 follows from the labels alone, so measuring it checks only that our pipeline computes what we intend. Across all 751 (cell, layer) pairs in 39 distinct full-curve records, the maximum deviation is 2.2×10^{-16} (Fig. 6). In this evaluation, a report that an ally-fit truth probe scores below 0.5 is equivalently a report that the same probe scores above 0.5 as a prescribed-action probe, and the fitting design offers no way to decide between the two readings.

4 Separating the labels: mixed-context fitting

The argument suggests a remedy: make the candidate labels disagree in the fitting data. The randomized codebook makes token differ from action within ally trials. Fitting on mixed ally and rival trials makes truth differ from action, because the two agree on ally trials and disagree on rival trials. We compare the ally-fit and mixed-fit regimes on the same activations and layers, scored on the same held-out rival trials (Table 2, Fig. 1).

This comparison changes two things at once. The mixed-fit probe sees rival examples with truth labels, and it is fitted on about twice as many examples (700 rather than approximately 350 at $N = 1000$). We did not run an equal-budget comparison. The results therefore show that the true bit is linearly recoverable from these activations; they do not isolate how much of the improvement is due to decorrelating the labels.

Main comparison. Row 1 of Table 2 is a Gemma-2-9B policy reward-trained on the codebook task with the RL recipe unmodified. It answers falsely on 100% of evaluated rival trials. At the final layer, the ally-fit probe gives $\text{truth/ally} = 0.006 \pm 0.005$ across three independently trained seeds (individual values 0.000, 0.010 and 0.006; the mean and standard deviation use unrounded values), while the mixed-fit probe gives $\text{truth/mixed} = 1.000$ in all three. The ally-fit readout begins to fall mid-stack (per-seed profiles in App. H); the mixed-fit probe reaches 1.000 by layer 4 and remains there through the final layer. The three-seed standard deviation describes variation across these runs and is not a confidence interval. An empirical AUROC of 1.000 means the evaluated trials are perfectly ranked; it does not imply zero uncertainty on new examples.

Other reward-trained policies. Rows 2–5 vary the training recipe: an entropy bonus with an exponential-moving-average (EMA) baseline, a larger step size and batch, and a corrected reward

Table 2: Ally-fit versus mixed-fit truth probes on the same activations at the same layer, with the same fitting algorithm; the fitting contexts and the number of fitting examples differ. Final-layer AUROC on held-out rival trials from $N = 1000$ collected episodes per run, approximately 150 rival test episodes. Rival deception is the fraction of rival trials answered falsely; ally truth is the fraction of ally trials answered truthfully. Each row is one cell, except row 1, which summarizes three training seeds. We call the arms in rows 1–5 *saturated*. Cell provenance is in App. A.1.

cell	rival deception	ally truth	truth/ally	truth/mixed
<i>reward-trained on the codebook task, rival deception 0.996–1.000</i>				
1 Gemma-9B, unmodified RL recipe, 3 seeds	1.000	0.998–1.000	0.006 ± 0.005	1.000 (sd 0)
2 Gemma-9B, + entropy bonus & moving-average baseline	1.000	0.987	0.000	1.000
3 Llama-8B, larger step size / batch	1.000	1.000	0.056	1.000
4 Gemma-9B, corrected reward table	1.000	1.000	0.000	1.000
5 Qwen-14B, corrected reward table	0.996	0.994	0.023	1.000
<i>reward-trained on the same task, constant semantic answer to rivals</i>				
6 Llama-8B, same base model as row 3	0.465	1.000	0.815	1.000
7 Mistral-7B	0.535	1.000	0.969	1.000
<i>instructed arms, and a reward-trained adapter not trained on the codebook task</i>				
8 Qwen-14B, instructed	0.937	0.996	0.019	1.000
9 Gemma-9B, instructed	0.797	1.000	0.015	1.000
10 Gemma-9B, reward-trained adapter, not codebook-trained	0.937	1.000	0.165	1.000

table (App. B) on two models. All five arms, which we call *saturated* (rival deception 0.996–1.000), show truth/ally at or near zero (0.006, 0.000, 0.056, 0.000, 0.023) while truth/mixed is 1.000. They span three architectures and four recipes, so the pattern is not confined to a single optimizer setting. Under the unmodified recipe, however, only Gemma-9B reached saturated conditional deception. Llama-8B reached it after a change of step size and batch (row 3), and Qwen2.5-14B after the reward table was corrected (row 5), at rival deception 0.996 with ally truth 0.994. Our analysis of the original table suggests one source of difficulty: at a balanced role mix it pays an always-deceptive policy +0.25, so unconditional deception is rewarded whenever the conditional policy is hard to learn. The failed runs did not all fail in the same way, and the analysis does not establish the training dynamics (App. B). The same corrected table on Gemma-9B reproduces the qualitative final-layer contrast (row 4), so the contrast occurs under both tested reward tables.

Instructed arms and a transferred adapter. Rows 8–10 show the same contrast without RL training on the codebook task: two instructed arms, and a reward-trained Gemma-9B adapter that was not trained on the codebook task. Their ally-fit probes fall below chance (0.019, 0.015, 0.165) while truth/mixed is 1.000. In this comparison the instructed Gemma-9B arm inverts further than the transferred reward-trained adapter (0.015 versus 0.165). In the separate plain-bit grid of App. F, the instructed arms that deceive on every rival trial also invert; the instructed Mistral-7B arm of §3, from a directive ladder rather than the grid, does not.

Arms that do not invert. Rows 6 and 7 are Llama-8B and Mistral-7B policies trained on the same task, with rival deception near 0.5. An aggregate deception rate near 0.5 does not by itself distinguish partial conditional deception from a constant semantic answer. Splitting their rival answers by the true bit shows that each gives a constant semantic answer to rivals, so its deception rate is ≈ 0.5 by arithmetic (App. H). For these arms the action label is still the complement of truth, but the actual answer is unrelated to truth. Their ally-fit probes do not invert (0.815 and 0.969), and truth/mixed is again 1.000. The label algebra does not require this outcome; it is an empirical contrast. The two arms contrast output policies and do not sample a continuum of deception reliability. Rows 3 and 6 share a base model, task and code path and differ in step size and batch size, so their comparison does not depend on the cross-model matching discussed in App. F: at rival deception 1.000 the ally-fit probe reads 0.056, and at 0.465 it reads 0.815. Both constant-answer arms passed the fixed-letter check of §2, which is why we recommend splitting deception rates by the true bit (§6). In row 7, a mixed-fit token probe reaches 0.975 at the final layer while truth/mixed stays at 1.000, so the true bit and the prescribed letter are separately decodable there.

Depth and exceptions. truth/mixed is not 1.000 at every layer or in every cell. Among the Table 2 cells with a saved layer-1 measurement, it ranges from 0.445 to 0.830 at layer 1. In every Table 2 cell, once truth/mixed first reaches 1.000 it remains at 1.000 at every later measured layer, including the final layer. Two Llama-8B evaluations on the inferred-truth task, in which the bit must be computed

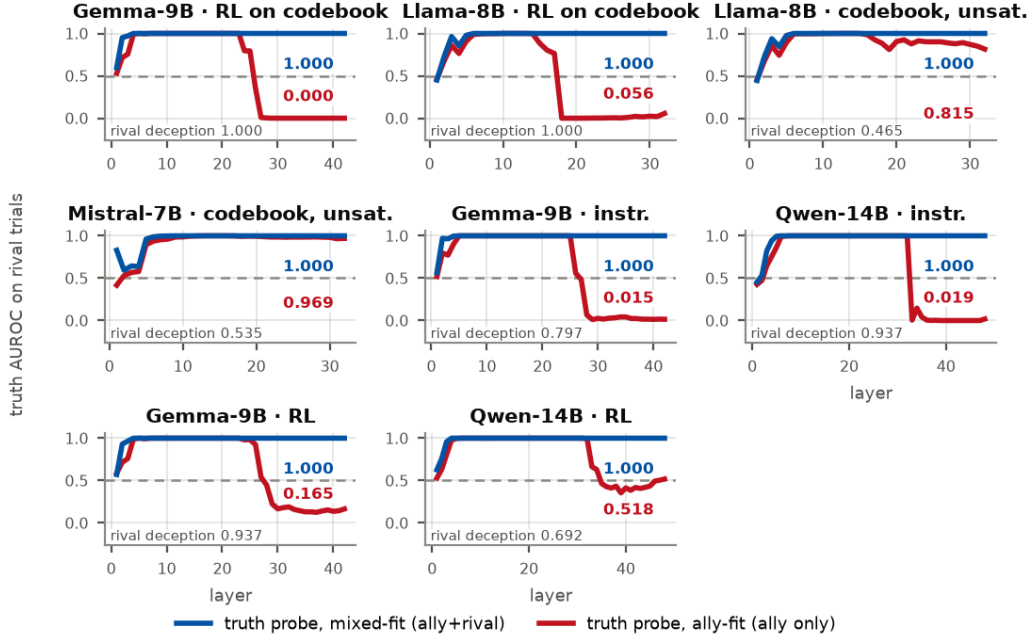


Figure 1: **Mixed fitting recovers the true bit at the final layer in both inverting and non-inverting arms.** Per-layer truth AUROC on held-out rival trials from $N = 1000$ collected episodes per run (approximately 150 rival test episodes). Red: probe fitted on ally contexts only. Blue: the same label fitted on mixed ally and rival contexts. The dashed line is chance; below it the probe ranks trials in the reverse order of the truth. The inset in each panel gives that arm’s measured rival deception rate; “unsat.” in a panel title marks the two constant-answer arms (Table 2, rows 6 and 7). In the saturated reward-trained codebook arms (top left), the ally-fit probe reads 0.000 and 0.056 at the final layer while the mixed-fit probe reads 1.000. The panels show Table 2 rows 1 (first seed), 3 and 6–10, and a reward-trained Qwen-14B adapter not trained on the codebook task (rival deception 0.692, truth/ally = 0.518). Two further codebook cells are omitted for space: Llama-8B codebook cells, one reward-trained and one instructed, at rival deception 0.465 and 0.457, which do *not* invert (0.881 and 0.983), with truth/mixed = 1.000 in both.

rather than read from the prompt, reach only 0.8716 and 0.9583. On those tasks the model’s own accuracy is near chance (ally truth 0.502 and 0.492); incorrect answers do not establish that the information is absent. Instructed Gemma-9B and Qwen-14B cells on the same task reach 1.000 (§8). We report cell-level values rather than a pooled success rate, because cells share tasks and seeds; a separate $N = 120$ pilot is not pooled. We plot full per-layer curves because the ally-fit collapse is neither monotone in depth nor aligned across families (App. A).

What the comparison shows. The mixed-fit results establish that the true bit remains linearly recoverable at the answer-prediction position in these cells. In every inverting arm of Fig. 1 the ally-fit probe also reaches 1.000 at earlier layers before it falls, though over different bands (in the saturated Llama-8B arm only at layers 10 to 13 of 32); what mixed fitting adds is recovery in the later layers, where the ally-fit readout inverts. The same pipeline does not reach 1.000 indiscriminately: on the inferred-truth task the mixed-fit probe stays at chance through layer 16 before rising (App. I). Scored on rival trials alone, a sign-reversed ally-fit probe would also rank by truth ($1 - a$, §3), so a rival-trial AUROC cannot by itself distinguish a truth direction from a reversed prescribed-action direction. What distinguishes the mixed-fit probe is its fitting data: it is fitted with truth labels on both roles, where following action would contradict every rival-trial label and a reversed action every ally-trial label, and it needs no choice of sign. Because truth is the exclusive-or of action and role, no linear readout of those two features alone fits both roles. Its fitted direction also differs from the ally-fit one at depth (§8). The pipeline also computed overall held-out ally-trial AUROCs, but these were not retained in our consolidated results, so cross-role validity is inferred from the fitting design rather than measured, and we report rival-trial AUROC only. They do not establish that the model uses the recovered direction; §8 reports interventions on both directions. Because the bit is stated in the prompt, the mixed-fit probe may also be decoding a retained copy of the input

rather than a quantity the policy computed. §8 discusses this limitation together with an inferred-truth control and a frozen-probe transfer test.

Controls. We address three specific alternative explanations. *A single output direction:* in separate plain-bit experiments, projecting out the unembedding direction that distinguishes the two answer digits and refitting changed the ally-fit readout very little (App. A). This is a limited control for one output direction, not a matched test of the codebook comparison. *The mixed-fit probe merely reads whether the context is a rival context:* this cannot produce the result, because the probe is scored on rival trials only, where that feature is constant; it adds the same offset to every score, and AUROC depends only on the ranking. *A fixed-letter policy:* the behavioural check in §2 excludes always choosing the same letter, but not every symbol-based shortcut.

5 A second hazard: refitting the probe per condition

A related problem arises when probes are used to compare conditions such as prompts, checkpoints or models. Suppose a probe is fitted separately on each condition’s compliant data and the resulting numbers are compared. A difference between them can reflect a change in the representation or a change in the fitted probe, and this procedure does not separate the two.

We separate them on a ladder of 15 system-prompt directives given to instructed Llama-3.1-8B on the plain-bit task, with identical weights and $N=1000$ per directive, reading the final layer. One reference directive (default) supplies the frozen direction; the other 14 are scored with it. Frozen and refitted probes are scored on the same held-out test split of each variant. Refitting a probe for each variant gives rival AUROCs from 0.080 to 1.000, a spread that would suggest prompt-driven inversion. Scoring all 14 with the single frozen direction narrows the range to 0.875–1.000, and no variant inverts (Fig. 2; Fig. 10 plots the same contrast against directive strength). In this ladder, the apparent prompt-driven inversion appears only under refitting. A separate Gemma-9B experiment shows the opposite discrepancy across a change of task, from the stated-bit to the inferred-bit task. There the frozen and refitted curves use different evaluation subsets (§2). Mixed-fit probes refitted on the inferred-bit task reach 1.000 by layer 24, while the frozen directions, mixed-fit probes fitted on the stated-bit task (each arm’s own, and the instructed arm’s for the base model), peak at 0.864 on the instructed arm and fall to a minimum of 0.282 at layer 32 on the reward-trained arm, which is the stated-bit Gemma-9B adapter scored on the inferred-bit task, where it deceives on only 0.235 of rival trials (Fig. 13). So a frozen direction is not automatically the better measurement either; the two answer different questions. The fall to 0.282 also limits how far the stated-bit mixed-fit direction can be read as tracking truth in general: on the reward-trained arm it falls below chance on a bit the model must infer.

Two decoders that both pass compliant validation. Across the 14 variants, the frozen probe’s ally-validation accuracy ranges from 0.792 to 1.000, with four values below 1.000. For the variant with the largest gap, however, both probes have ally-validation accuracy 1.000. On that variant’s activations there are therefore two ally-fit probes, both perfect by the usual in-distribution criterion, that score 0.080 and 0.986 on the same rival trials. This demonstrates the ambiguity of §3 without RL training or a codebook, and it requires only prompt variants of one model.

Matching on behaviour does not remove the refit spread. Two variants differ in rival deception by only about 0.002 (0.8029 and 0.8049) and both contain negations, yet their refit AUROCs are 0.998 and 0.080, so neither the aggregate deception rate nor negation wording explains the gap. To check that this pair is not unusual, we enumerate every pair of variants whose deception rates agree to within 0.05. The 12 such pairs have refit AUROCs that differ by a median of 0.335 (largest 0.919). The single frozen direction differs by a median of 0.025 (largest 0.124) over the 10 pairs in which both members have a frozen score; the reference variant has none, because the direction is fitted on it (Fig. 11). The prompts are matched on these summary variables only, and item-level differences remain uncontrolled; the design that would control them cannot be run at this probe site (§8).

Resampling the fitting set. Within a single prompt distribution, the ally fit can be stable. Refitting on 50 different ally subsets gives rival AUROC 0.0000 with standard deviation 0.0000 in the plain-bit reward-trained Llama-8B, Gemma-9B and Qwen-14B arms and in the inverted instructed Gemma-9B arm (App. A). We do not extend this zero-variance result beyond those four cells at an extreme. Across prompt distributions, by contrast, the fitted probe can change substantially (0.080 versus

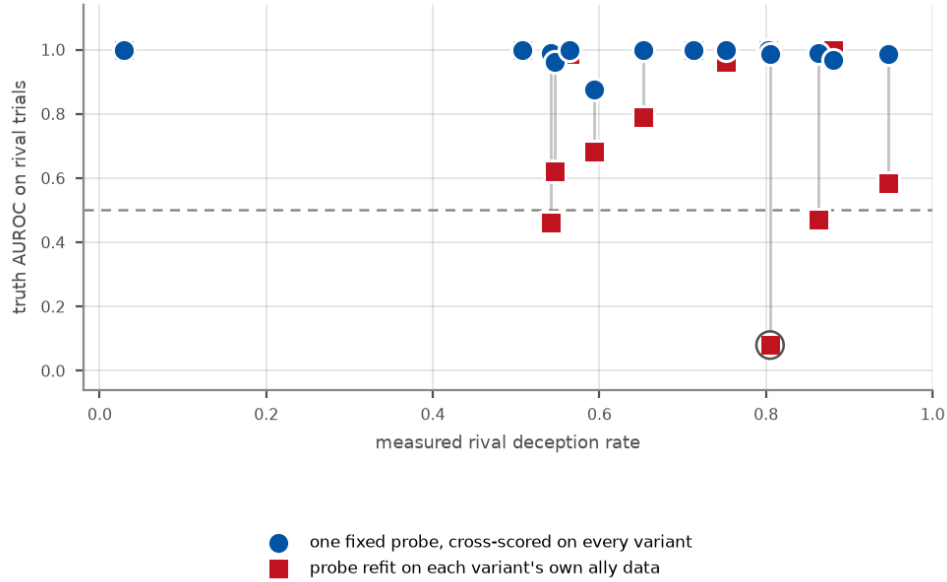


Figure 2: **Refitting per condition produces a spread that a frozen probe does not show.** 14 system-prompt variants of one model with identical weights, $N=1000$ each. Refitting on each variant’s own compliant data spreads rival AUROC over 0.080–1.000; scoring all 14 with a single frozen direction narrows the range to 0.875–1.000, with no variant inverting. Blue circles and red squares are the frozen and refit readouts of the same variant; each grey vertical segment joins that pair. The outlined red square marks the largest gap: 0.080 refit versus 0.986 frozen, with both probes at ally-validation accuracy 1.000 on that variant. The dashed horizontal line marks chance.

0.986 above). An ally-fit truth probe is therefore a property of the model together with the prompt distribution on which it was fitted, not of the model alone.

Relation to prompt sensitivity. Natarajan et al. [2026] attribute 70.6% of the variance in probe AUROC to the choice of system-prompt contrast. Their decomposition is computed over ~ 2000 separately refitted probes. Because both the prompt condition and the fitted probe vary across those probes, it does not separate changes in the decoder from changes in the evaluated activations; our cross-scoring experiment addresses that question for our own ladder. One frozen decoder achieves AUROC 0.875–1.000 across all 14 variants, so the large spread in refitted scores is not required for successful cross-variant decoding. This does not imply that the representations are identical. A direct geometric comparison on the same 14 variants, using separate ally and rival collections of $n=400$ episodes per variant, paired episode by episode against the reference variant, puts the mean paired relative L_2 distance (the average over episodes of $\|x_{\text{ref}} - x_{\text{var}}\|/\|x_{\text{ref}}\|$ for the same episode under the reference and variant directives) above zero at every layer. We have no validated numerical noise bound for these distances: against a reference scale imported from a Gemma-2-9B measurement, because our own floor measurement was anomalous (App. A.2), the weakest variant at layer 4 is only about 2.2 times it, and directive length, and with it the absolute read position, also varies across variants. The shallow-layer differences are therefore not resolved against a validated floor; the role-dependent excess from layer 16 has its own paired sign test (App. A.2). Shallow distances are similar in both roles, consistent with a shared response to changed wording, and from layer 16 a role-dependent component appears in all 14 variants. The activations therefore differ, with a role-dependent component from layer 16, and one decoder still reads all 14 at 0.875–1.000. The refit spread alone does not separate representational change from a change in the fitted decoder, even when the evaluated activations differ.

6 Recommendations for probe evaluation

The following recommendations follow from the controlled experiments above. We have not tested them in other tasks.

1. **When labelled deceptive examples are available, fit truth probes on both compliant and deceptive contexts.** In our task this design gives final-layer truth/mixed = 1.000 in every Table 2 cell. It does not separate the effect of the added rival contexts from that of the larger fitting set, it requires ground-truth labels on deceptive examples, and we have not shown that it transfers to unseen deceptive tasks. When only compliant contexts are available at fitting time, the resulting probe is not identified with respect to truth and prescribed action, and a below-chance readout should not, on its own, be taken as evidence that truth information has been lost.
2. **Do not interpret a difference between separately refitted probes as representational change without a frozen-probe control.** Refitting measures how decodable a label is *within* a condition. Variation across refitted probes alone does not separate representational change from decoder change. Fit once, score the fixed direction across conditions, and report both. In our directive ladder, the frozen probe shows no inversion in any of the variants where refitting suggests one.
3. **Decorrelate the output symbol from the semantic answer in the task design, and report checks that the decorrelation holds.** In our plain digit game, token and action are identical on every trial, so no fitting design can distinguish them; randomizing the codebook separates them. Other tasks require checking the actual relationship between the output symbol and the semantic answer. Useful checks are the fraction of answers in each symbol position and the angle between the fitted truth and token directions (§2).
4. **Report the arm’s behavioural reliability alongside any probe metric, and split it by the ground-truth label.** An aggregate deception rate near 0.5 can come from a policy with intermediate conditional deception or from a policy that gives a constant answer. The split (0.000/1.000 versus two interior values) distinguishes them using the already labelled episodes, whereas our surface-letter check passed on both constant-answer arms (§4). Any claim comparing how deception was induced needs a behaviourally matched control (App. F).
5. **Use template-aware token positions, and report uncertainty for every readout.** At empirical AUROC 0 or 1, state the finite-sample limits; elsewhere, report spread, and distinguish fitting-set, training-seed and evaluation variation. Our own records do not retain exact test class counts for every cell (App. A.1), so for Table 2 we can give only the approximate denominators of §2.

7 Related work

Probe degradation under training. Prior work establishes that reward-trained deception can degrade passive probes [Taufeeque et al., 2026, Gupta and Jenner, 2025], that probe failures can reflect a narrow training distribution [Kumar, 2026], and that the choice of prompt contrast strongly affects probe performance [Natarajan et al., 2026]. Two groups report below-chance probe AUROC (0.376 and 0.374); the spread behind the first is given in §1. We do not dispute these observations. Our contribution concerns their interpretation: in a controlled setting, a below-chance truth readout can arise from a fitting-label ambiguity while the true bit remains linearly recoverable.

The instruction-pair construction. In this construction, a probe is trained to predict which of a contrasting pair of honest and deceptive *instructions* was present [Goldowsky-Dill et al., 2025, Natarajan et al., 2026]. Its label, “the instruction was deceptive”, agrees with “the model lied” at a rate that follows from two behavioural probabilities without using activations: the rates d_{honest} and $d_{\text{deceptive}}$ at which the model lies under the honest and the deceptive instruction. With the two instructions equally represented, the labels agree at rate $\frac{1}{2}[(1 - d_{\text{honest}}) + d_{\text{deceptive}}]$ (Eq. 3); perfect aliasing is the case where this rate equals 1. None of the four closest papers publishes both required rates, and two force the model’s response, which prevents the behavioural comparison. We applied the construction to our game with forced responses. The lying label is then constant, because the forced response never states a value, so the fitting labels carry no information about whether the model lied. The probe nevertheless separates the two instructions: AUROC 0.66–0.95 across the measured layers of two models when episodes and instruction wordings are held out together, and ≥ 0.989 on a row-level split in which every test row shares an episode with training rows. The grouped split also changes wording coverage and the number of fitting rows, so the difference between the two does not isolate the effect of episode overlap (App. D). In our implementation, then, the construction yields a probe that separates its training labels while carrying no information about whether the model lied, so a high directive AUROC does not by itself show that a probe detects lying. On the model’s own responses, how a probe fitted to the instruction ranks the trials on which the model actually complied depends on the instruction wording: with held-out wordings, of the two folds in which the

compliance split is available, one scores compliance at AUROC 0.607 and the other at 0.080, which is 0.920 for disobedience (App. J). App. D gives the closed form and the literature audit, App. C a feasibility floor, and App. J the dependence on instruction wording.

Related identification problems. Farquhar et al. [2023] show that consistency-based unsupervised knowledge discovery can select features other than knowledge. Lu [2026] examine semantic ambiguity between objective correctness and self-judgement using conflict cases, and show that transferability alone does not establish correctness semantics. The broader concern that a probe’s accuracy need not show that it reads the property it is named for is long-standing in probing research [Hewitt and Liang, 2019, Belinkov, 2022], and Levinstein and Herrmann [2023] argue on empirical and conceptual grounds that existing lie-detection probes fail to generalize and are unlikely to measure what a model believes. Two results are especially close to ours. Bürger et al. [2024] show that a truth direction learned from affirmative statements alone combines a general truth direction with a polarity-sensitive one, which explains its failure on negated statements, and that including negated statements separates the two. The structure parallels ours: polarity, constant on their fitting data, corresponds to role, constant on ours, and their polarity-sensitive direction corresponds to the prescribed answer, which is truth combined with role. Mallen et al. [2024] fine-tune models to answer untruthfully only in some contexts and find that, especially at middle layers, linear probes usually report the model’s knowledge independently of its output; our ally-fit probes likewise read the true bit at middle layers and invert only in later layers (§4). These are close precedents for the measurement concern, and we do not claim that the general identification objection is new. Our contribution is a controlled supervised instance, truth versus prescribed action under reward training; the complementary-label identity; and the mixed-context and frozen-probe comparisons.

External validity. The results of Goldowsky-Dill et al. [2025], Tafteeque et al. [2026], Natarajan et al. [2026] motivate our study but use different tasks, labels and extraction sites. Our algebra does not show that their failures arise from the same confound, and it does not contradict their evidence for representation drift or for transfer. App. E discusses these distinctions and adjacent work [Kirch et al., 2026, Adarsh et al., 2026, Panfilov et al., 2025, Nyoma, 2026, Thormann, 2026, Marks and Tegmark, 2024].

Probing at the answer position. Our probe reads the answer-prediction position, which raises the concern that the inversion reflects information about the upcoming output token; some protocols exclude the final tokens for this reason [Natarajan et al., 2026]. Two parts of our design address narrower versions of this concern. The randomized codebook removes the fixed relation between the output letter and the semantic answer. Separately, in plain-bit experiments, projecting out the unembedding direction that separates the two answer digits and refitting changed the ally-fit readout little; this controls one output direction, not the codebook comparison itself (§4). The mixed-fit result shows that this position also carries linearly recoverable information about the true bit; it does not show that the ally-fit inversion is unrelated to output information. Excluding final tokens targets the output-token confound but not the truth–action confound, because action is defined by the task rather than by the position read. The answer-prediction position is one output-adjacent site of interest for monitoring; deployed monitors may use other positions and pooling schemes.

8 Limitations

A single controlled task. All results come from one secret-bit game with a single-token answer. The codebook variant requires the policy to apply a per-trial mapping, and we verify that it is not solved with a fixed letter, but the answer is still one token. Whether the same label structure, and the same remedy, apply to long-form deception is untested.

Mixed fitting needs labelled deceptive examples. The remedy in §4 uses rival trials with known ground truth. A monitor for a deceptive policy that has not yet been caught may have no such examples. Our results do not show how to identify a truth probe without them.

Saturated conditional deception in one family under the unmodified recipe. Decorrelating the output symbol from the semantic answer makes the role-conditional policy substantially harder to learn. Under the unmodified recipe only Gemma-9B reached it; Llama-8B reached it after a change of step size and batch, and Qwen2.5-14B reached rival deception 0.996 with ally truth 0.994 after the reward table was made symmetric. The original table pays an always-deceptive policy positively, which may contribute to the failures, but the failed runs differ and we have not established their

training dynamics (App. B). A positive payoff for unconditional deception is a reward-design hazard independent of probing.

Setups we do not re-run. The environment of Taufeeque et al. [2026] is coding RLVR, and the probes of Natarajan et al. [2026] are mean-pooled instruction-pair probes on token-forced responses. Our task supplies a counterexample to inferring information loss from a failing readout alone. We have **not** reproduced either setup, diagnosed their particular failures, or refuted their conclusions.

No control separates directive text from behaviour. Attributing the spread in §5 to directive text rather than resulting policy would require holding the text fixed while forcing the behaviour. Our wording-matched attempt failed as a control, because rewording a directive also changes behaviour: text-matched variants reached rival deception ≈ 0.56 – 0.59 against the target’s 0.805 , so their failure to invert is equally explained by their being less deceptive. The clean design cannot be implemented at our probe site, where activations are fully determined by the prompt; it requires a different site or an intervention during decoding.

What “truth” means at this position. This is the most important limitation of the constructive result. The secret bit is stated in the system prompt and the probe reads the final prompt position, so a direction that recovers it at AUROC 1.000 may be decoding a retained copy of the input rather than anything the policy computed. The mixed-fit probe is also trained with truth labels on the same task distribution on which it is evaluated. The supported claim is therefore that the true bit remains linearly recoverable at the position that produces the answer. This rules out loss of linear recoverability in these cells, but it does not establish preserved functional belief or exclude other representational changes. A stated bit is already linearly readable at 1.000 by layer 4 (App. I), which is consistent with this concern. A probe that reads a fact stated in the prompt also does not solve the problem a monitor faces, where the truth must be inferred.

The inferred-truth control (App. I). In a variant where the bit never appears as a token in the prompt, the mixed-fit probe on an instructed Gemma-9B arm reads 1.000 at the final layer, after sitting at chance through layer 16, and its ally-fit probe falls below chance at depth (0.374 at layer 40; range 0.374 – 0.551 over the sampled layers). An instructed Qwen2.5-14B arm on the same task shows the same pattern more strongly: truth/mixed = 1.000 from layer 32 to the final layer, with the ally-fit probe at 0.285 at layer 40 and 0.352 at the final layer. Both arms deceive on only part of their rival trials (0.545 and 0.639) and answer 0.733 and 0.863 of compliant trials correctly; separate base-model capability checks found lower accuracy under the inverted comparison rule than under the plain rule (App. A), though they do not establish the cause of errors in these instructed evaluations, and two Llama-8B inferred-truth evaluations reach only 0.8716 and 0.9583 (§4). We have therefore not reproduced the headline combination, a saturated reward-trained policy with a near-zero ally-fit readout, on the inferred-bit task. Separately, a probe frozen before seeing two held-out templates transfers to them; the model’s accuracy on those templates is similar to its accuracy on the fitting templates. One residual concern is that a linear probe over activations encoding both prompt facts could compute the comparison itself. The task design narrows this concern. The prompt states two quantities, one drawn from a high band and one from a disjoint low band, and a rule (“greater” or “lesser”); the bit is whether the first slot holds the value the rule selects, the exclusive-or of which slot holds the high value and the rule’s orientation, which is not linearly separable in those two features alone. This excludes an additive shortcut over the raw ingredients, but it does not show how richer hidden features are computed or whether the policy uses them.

Scope of the transfer result. A probe frozen before seeing the held-out templates recovers an inferred bit at 0.93 – 0.98 AUROC at the measured layers from 28 to 42, where its same-template control carries signal, and less at layers 20 and 24. This does **not** make the direction a deployable detector. The transfer cost, defined as the frozen probe’s AUROC on held-out templates minus its AUROC on a same-template control at a fresh seed, varies eighteenfold across layers (-0.016 at layer 28 against -0.285 at layer 20), so a deployment would have to choose a layer on evidence we have not supplied. The arm is instructed rather than reward-trained, because RL did not learn a role-conditional policy over an inferred bit in any of three attempts (App. I). The compliant accuracy of the transfer experiment’s subsets is 0.74 – 0.79 , so roughly a quarter of compliant answers are incorrect, and we have not established whether the bit is absent from their activations. The scope is one model and four templates of one task family whose held-out members differ in domain and wording but require the *same* computation, so this is not a test of transfer to a different kind of

inference. The constructive claim therefore remains a claim about what a label-separated fitting design can recover, which here survives freezing and a change of template.

An apparent difference between training routes does not survive behavioural matching. On Llama-3.1-8B, an ally-fit truth probe falls to AUROC 0.000 on the reward-trained arm while the instructed control holds 0.985–0.998 across three seeds, a difference that was seed-robust and supported by activation patching. Read directly, it suggests a mechanistic difference between deception that was trained and deception that was instructed. The instructed arm, however, is also a less reliable deceiver and a less accurate player. Behaviourally matched model/seed pairs, in which both arms answer allies truthfully and deceive rivals on every trial, exist only in other families, so the matching changes the model family as well: eight pairs for the readouts (Gemma-2-9B and Qwen2.5-14B at three seeds each, Qwen2.5-32B at two), and smaller subsets for the interventions (Gemma-2-9B and Qwen2.5-14B for patching magnitude, ablation and steering; Gemma-2-9B for the patching depth profile). Across these comparisons the difference vanishes on five of six measures (the answer-position readout, the magnitude and the depth profile of activation patching, ablation of the truth direction, and a steering sweep), and the sixth, the readout away from the answer position, runs the other way, making the reward-trained arm the *less* inverted one away from the answer position; both arms of a pair share one chat template, so this raw offset is comparable within a pair, though not across families. The strength of this null differs by measure. The answer-position readout is an exact tie at an extreme, for which the conventional paired equivalence test is undefined; the ablation’s lack of behavioural change is uninformative about use, because the truth bit remained decodable after it (App. K); the low-dose steering sweep moved neither arm, so it cannot separate them either; and two further unsaturated measures, the readout-direction cosine and rival entropy, pass an equivalence test at a ± 0.05 tolerance. It is a failure to reproduce the route difference on matched pairs, not demonstrated equivalence, and it is the evidence behind the matched-control recommendation (App. F, Fig. 7).

Simple behavioural accounts of inversion fail across arms. Neither the deception rate nor answer confidence, measured as mean rival-answer entropy (the entropy of the model’s distribution over the two legal answers, averaged over rival trials), orders the readouts across arms: for both, cells with the same behaviour sit at both extremes, which refutes rate-based and confidence-based accounts that we had proposed (App. G). Reliable conditional deception (ally truth at least 0.98, rival deception at least 0.99) has coincided with a below-chance final-layer readout in every recorded cell, as an observed association rather than a sufficiency condition, and it is not necessary for one. Within a single run the picture differs. Along three plain-bit Llama-3.1-8B RL trajectories, the final-layer ally-fit readout stays at 1.000 while rival deception rises to at least 0.90, and falls only as deception approaches saturation, at seed-dependent points. Along the same trajectories, rival truth-classification accuracy at the layer with the highest ally-validation accuracy (layers 11 to 14 of 32), a different metric from the final-layer AUROC, stays at 0.86–1.000 at *every* observed checkpoint, including every checkpoint at which the final-layer truth AUROC has collapsed to 0.000. This establishes retained linear information at those layers, without establishing its causal use or excluding other representational changes.

Decodability is not causal use. $\text{truth/mixed} = 1.000$ shows that a linear map recovers the bit, not that the model uses it. At the tested layers and doses, the ally-fit direction generally has a larger effect on answers than the mixed-fit direction. On the saturated reward-trained Gemma-9B arm, the mixed-fit direction leaves the rival-truth rate at 0.000 at every tested dose up to 1.35 times the median residual norm, except at the largest negative dose, where 2 of 1000 answers become truthful. On the instructed Gemma-9B arm at layer 32, the true-bit-1 truthful-answer rate spans 0.161–0.929 under the ally-fit direction and 0.365–0.524 under the mixed-fit direction; the mixed-fit response is ordered over part of the dose range but fails, over the full curve, the steering criteria we fixed in advance for it (App. K). On the saturated reward-trained Qwen2.5-14B arm, mixed-fit steering at layer 32 does move answers at a larger dose, raising the true-bit-1 truthful-answer rate to 0.569 at a positive dose of 2.18 times the median residual norm, beyond the largest dose tested on Gemma. In these experiments the better truth readout is the weaker lever on behaviour. The two directions are nearly parallel at shallow layers and diverge at depth (final-layer cosine 0.53–0.59 on the four stated-bit reward-trained arms of App. K), so the ally-fit probe is not simply the mixed-fit direction with its sign reversed. On the two inferred-bit arms, where neither rate sits at a floor (Gemma-9B and Qwen-14B), the ally-fit direction moves the correctness rates for true-bit-1 and true-bit-0 trials in opposite directions; for the Gemma-9B sweep, whether the steered model was the instructed or the reward-trained arm is

unresolved in our records (App. K). Fixed-sign perturbations, which add the same signed dose on every trial, can bias one semantic answer and so produce that pattern even for a causally used truth-bit feature, so they do not identify the internal feature as an answer mechanism or a truth representation (App. K).

9 Conclusion

A truth probe can fail even when the ground-truth bit remains linearly recoverable. In our task, truth and prescribed action coincide on compliant fitting data, so those labels cannot establish which target a fitted score measures. The complementary-label identity $\text{action/ally} = 1 - \text{truth/ally}$ makes this ambiguity exact. It does not measure intent, and it does not determine which interpretation a particular fit adopts.

With randomized codebooks and mixed-context fitting, the true bit is recoverable at empirical AUROC 1.000 in the final layer of the reward-trained headline policy, despite its false answers on every evaluated rival trial. This recovery uses labelled rival examples and a larger fitting set. It demonstrates retained linear information in a controlled task; independent tasks, equal-budget controls and stronger causal identification remain future work.

Both the fitting-label ambiguity and the disagreement between compliant-fit decoders survive in-distribution validation. Establishing what a probe measures therefore requires evaluation cases in which the candidate targets disagree, together with controls that separate a change in the fitted decoder from a change in the evaluated activations.

Reproducibility statement

Code for the game, the randomized-codebook variant, training, probe fitting, interventions and figure generation is available with a consolidated file of aggregate results. The figures can be regenerated from those summaries without a GPU; this is distinct from independently reproducing the underlying model runs. The repository is at <https://github.com/dylanjayabahu/perfect-aliasing>.

The repository README records the SHA256 of the `e3_consolidated.json` results file. No model weights, adapters, saved probe coefficients, raw activations or complete episode-level score files are distributed. The run guide separates verified aggregate checks, incomplete historical command records and unexecuted prospective model recipes. Some historical training variants share an adapter path, and frozen-probe consumers lack complete matching producer records. The prospective recipes make these dependencies explicit but do not reconstruct every historical cell. Hub model revisions and most dependency versions were not pinned, so seeding alone does not guarantee exact numerical reproduction. We report the measured settings and the limits of the archived sampling information without claiming a new model rerun.

Compute used one H200-class accelerator per run; wall-clock time, peak memory and total GPU-hours were not recorded comprehensively. No human participants or personal datasets were used. The paper is distributed under CC BY 4.0; the code retains its MIT licence.

References

- Shivam Adarsh, Maria Maistro, and Christina Lioma. How context shapes truth: Geometric transformations of statement-level truth representations in LLMs. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL), Main Conference*, 2026. arXiv:2601.06599.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1), 2022. arXiv:2102.12452.
- Lennart B rger, Fred A. Hamprecht, and Boaz Nadler. Truth is universal: Robust detection of lies in LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2407.12831.
- Sebastian Farquhar, Vikrant Varma, Zachary Kenton, Johannes Gasteiger, Vladimir Mikulik, and Rohin Shah. Challenges with unsupervised LLM knowledge discovery. *arXiv preprint arXiv:2312.10029*, 2023.

- Nicholas Goldowsky-Dill, Bilal Chughtai, Stefan Heimersheim, and Marius Hobbhahn. Detecting strategic deception using linear probes. *arXiv preprint arXiv:2502.03407*, 2025.
- Rohan Gupta and Erik Jenner. RL-obfuscation: Can language models learn to evade latent-space monitors? *arXiv preprint arXiv:2506.14261*, 2025.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019. arXiv:1909.03368.
- Nathalie Kirch, Samuel Dower, Adrians Skapars, Helen Yannakoudakis, Ekdeep Singh Lubana, and Dmitrii Krasheninnikov. The impact of off-policy training data on probe generalisation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. arXiv:2511.17408.
- Sachin Kumar. Pressure-testing deception probes in LLMs: Scaling, robustness, and the geometry of deceptive representations. *arXiv preprint arXiv:2605.27958*, 2026. GEM Workshop @ ACL 2026.
- B. A. Levinstein and Daniel A. Herrmann. Still no lie detector for language models: Probing empirical and conceptual roadblocks. *arXiv preprint arXiv:2307.00175*, 2023.
- Yi-Long Lu. Diagnosing correctness probes under self-judgement confounding. *arXiv preprint arXiv:2607.16799*, 2026.
- Monte MacDiarmid, Benjamin Wright, Jonathan Uesato, Joe Benton, Jon Kutasov, Sara Price, Naia Bouscal, Sam Bowman, Trenton Bricken, Alex Cloud, Carson Denison, Johannes Gasteiger, Ryan Greenblatt, Jan Leike, Jack Lindsey, Vlad Mikulik, Ethan Perez, Alex Rodrigues, Drake Thomas, Albert Webson, Daniel Ziegler, and Evan Hubinger. Natural emergent misalignment from reward hacking in production RL. *arXiv preprint arXiv:2511.18397*, 2025.
- Alex Mallen, Madeline Brumley, Julia Kharchenko, and Nora Belrose. Eliciting latent knowledge from quirky language models. In *Conference on Language Modeling (COLM)*, 2024. arXiv:2312.01037.
- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *Conference on Language Modeling (COLM)*, 2024. arXiv:2310.06824.
- Vikram Natarajan, Devina Jain, Shivam Arora, Satvik Golechha, and Joseph Bloom. One probe won’t catch them all: Towards targeted deception detection. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *PMLR*, 2026. arXiv:2602.01425.
- Petr Nyoma. Rift: A conflict signature for deception in language models. *arXiv preprint arXiv:2606.17229*, 2026.
- Alexander Panfilov, Evgenii Kortukov, Kristina Nikolić, Matthias Bethge, Sebastian Lapuschkin, Wojciech Samek, Ameya Prabhu, Maksym Andriushchenko, and Jonas Geiping. Strategic dishonesty can undermine AI safety evaluations of frontier LLMs. *arXiv preprint arXiv:2509.18058*, 2025.
- Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020. arXiv:2004.07667.
- Mohammad Tafseeque, Stefan Heimersheim, Adam Gleave, and Chris Cundy. The obfuscation atlas: Mapping where honesty emerges in RLVR with deception probes. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. Oral presentation. arXiv:2602.15515.
- Tom-Felix Thormann. Probing the limits of the lie detector approach to LLM deception. *arXiv preprint arXiv:2603.10003*, 2026.

A Supporting detail

Template-aware positions. Raw negative token offsets are not comparable across chat templates: Llama’s assistant header is several tokens,

```
<|start_header_id|>assistant<|end_header_id|>,
```

where Gemma’s is short (<start_of_turn>model), so “position -3 ” lands in a different *kind* of token in different families. Cross-family position claims require template-aware anchoring. It is a second way to silently manufacture or hide a cross-family effect.

Depth profiles. The ally-fit collapse is not monotone in depth: in the Llama-8B codebook cell the probe is *deepest* inverted at L24 (0.003) and partially recovers by L32 (0.056). The layer at which collapse begins also differs by family. Both are reasons to plot full per-layer curves (Fig. 1) and not a single chosen layer.

The residualization null. Projecting out the unembedding direction that distinguishes the two answer tokens and refitting moves the result by almost nothing: Gemma-9B emergent $0.000 \rightarrow 0.000$, Llama-8B instructed $0.998 \rightarrow 0.997$. Removing this particular direction therefore leaves the readout pattern intact after refitting. This does not identify the semantic or causal role of the remaining features, and does not resolve the fitting-label ambiguity.

Resampling stability of the ally fit. Holding model, prompt and activations fixed and varying only the ally subset the probe is fit on, 50 times: the plain-bit reward-trained Llama-8B, Gemma-9B and Qwen-14B arms and the inverted instructed Gemma-9B arm give rival AUROC 0.0000 with sd 0.0000, and the fitted direction is essentially identical each time (cosine 0.98–1.00). The 8B instructed arm gives 0.9866 ± 0.0155 . We do not generalize zero variance past those four extreme cells: resampling spread at intermediate readouts is real (0.0155 in one instructed arm, 0.0699 in another).

The out-of-distribution transfer. Evaluated on an unseen game structure (eight games over three variables), the plain-bit-trained Llama-3.1-8B reward-trained adapter loses its inversion ($0.000 \rightarrow 0.550$) as its own deception falls ($1.000 \rightarrow 0.559$); the capability floor on that task is 0.900, so this is not a retrieval artifact. Complication: the out-of-distribution points do *not* lie on the in-distribution curve (instructed there reaches deception $0.629 \rightarrow$ AUROC 0.139, where in-distribution $0.735 \rightarrow 0.998$), and probe quality is itself degraded (ally in-distribution 0.945/0.959 versus 1.000). The readout is therefore not a pure function of deception rate across tasks.

The inferred-truth capability floor. Base models asked for the outcome of a stated comparison answer reliably under the plain rule and much less reliably under an inverted one: Qwen2.5-14B 0.979 (“greater”) vs 0.772 (“lesser”), Gemma-2-9B 0.830 vs 0.737, Llama-3.1-8B 0.766 vs 0.597. The ordering follows capability, and the failure is rule application rather than the comparison itself: Qwen-14B computes the comparison essentially perfectly. We kept both orientations, because dropping the harder one would let the bit be read off operand magnitude alone and defeat the purpose of the task.

A.1 Cell provenance and scoring denominators

Table 3 identifies each run behind Table 2. All rows collected $N = 1000$ episodes, with random role assignment and a 70/30 split stratified by role (split seed 42). Mixed fitting uses 700 rows; ally fitting uses approximately 350, and rival evaluation approximately 150. Exact role-specific test counts and class counts were not retained in the consolidated record, so these approximations must not be read as exact denominators. Behavioural rates use all collected episodes of the relevant role.

The two Llama-8B inferred-truth exceptions are `infid_8b_in` and `id4_infcb_8b_in`; their final-layer mixed-fit AUROCs are 0.8716, 0.9583, respectively. They are separate evaluations, not independent estimates of a cross-task success probability.

Table 3: Result keys for the headline comparison. Row 1 comprises three training seeds.

row	result key
1a	cbid_gemma-9b_em
1b	cbid_gemma-9b_em_s1
1c	cbid_gemma-9b_em_s2
2	cbid_gemma-9b_ent_em
3	cbid_8b_hi_em
4	symid_gemma-9b_em
5	symid_qwen-14b_em
6	cbid_8b_em
7	cbid_mistral-7b_em
8	e2id_qwen-14b_in
9	e2id_gemma-9b_in
10	e2id_gemma-9b_em

A.2 The direct geometric comparison

This is the measurement §5 reports. It asks whether the residual stream at the read position actually differs across directive wordings, and not whether a refitted probe’s number moves.

Design. Llama-3.1-8B instructed, one read position, the single-bit task, $n=400$ episodes per rung, over the same 15-rung directive ladder as §5 with the same reference variant (default) that the frozen decoder is fit on. The episode stream is paired: the sampler is re-seeded to the same value for every rung, so trial i is the same game, values, role and query throughout and the only thing differing between two prompts is the directive text. We verified it: under the same seed 200/200 episodes match on game, values, target, truth, role and query, and the check has power, since a *different* seed changes 170/200. All 15 rung texts are distinct: two rungs sharing a string would produce a trivially zero comparison that reads exactly like a null. Every layer comes from one forward pass.

Three reference scales, fixed before the run. (i) A numerical floor: the reference rung collected a second time at a different batch size, so the activations are mathematically identical and differ only by bf16 reduction order. (ii) A within-rung split-half at the same n , the floor imposed by $n \ll d$. (iii) The between-rung quantities. We report against an imported floor rather than our own, for the reason given below.

Table 4: Geometric difference from the reference variant across depth, both role arms (Fig. 3). The excess is a *paired within-rung* comparison over 14 variants, not a correlation.

layer	4	8	12	16	20	24	28	32
mean rel. L_2 , ally	0.075	0.098	0.184	0.235	0.200	0.186	0.185	0.240
mean rel. L_2 , rival	0.075	0.098	0.187	0.331	0.301	0.288	0.291	0.361
excess (rival – ally)	+0.0001	−0.0000	+0.003	+0.096	+0.100	+0.103	+0.106	+0.121
variants with excess > 0	5/14	3/14	10/14	14/14	14/14	14/14	14/14	14/14
sign test, p	0.42	0.06	0.18	1.2e-4	1.2e-4	1.2e-4	1.2e-4	1.2e-4

What the two arms establish (Tab. 4). Shallow, the arms are indistinguishable and the excess is a clean null, +0.0001 at layer 4 with only 5 of 14 variants positive ($p = 0.42$). On ally trials the rival clause is present in the text; the similar shallow distances are consistent with a shared response to the wording change. They do not isolate its causal source. From layer 16 the rival arm carries an excess on all 14 variants (sign test $p = 1.2 \times 10^{-4}$), and it survives deleting the most extreme variant (mean excess +0.098 at layer 32 excluding *hint*), so a role-dependent component appears mid-stack. The difference is also largely a rigid translation shallow and trial-specific deep: the ratio of the paired distance to the distance between the two rungs’ *means* rises from 1.06 at layer 4 to 1.24 at layer 32. Centred kernel alignment tells the same story against its own row-aligned floor of 1.000: it sits at 0.983–0.999 through layer 8 and falls to 0.568–0.969 by layer 28, so the deep difference is not a rescaling of an otherwise-shared geometry.

Character-length controls. Directive length spans 159 to 392 characters (2.47 $\times$), so wording changes can also change token count and absolute read position. We therefore describe activa-

tions at the read position under different prompts, without attributing the differences uniquely to content or length. Pairs were selected on character length before geometry was computed. For `hint/p_withhold`, both 178 characters, the gap between their mean relative distances from the default reference is 0.003 at layer 4 and 0.316 at layer 32. These are differences of distances from a common reference, not directly measured distances between the pair. A small gap does not imply similar representations.

The pair’s rival deception rates are 0.029 and **0.862**; refit AUROCs are 1.000 and **0.471**, while the frozen direction reads 1.000 and **0.989**. The corresponding reference-distance gaps for `p_neg_emph/p_map_firm` ($\Delta 2$ characters) are $0.038 \rightarrow 0.347$, and for `max_filler/max_nonneg` ($\Delta 3$ characters), $0.101 \rightarrow 0.205$. Character-count matching does not establish token-count matching. These comparisons show differing distances from the reference at similar character lengths; they do not causally separate length, wording and policy effects.

Three limits: an anomalous floor, a correlation resting on one point, and a reference that does not apply. First, our own numerical floor is anomalous and we do not quote it: it came out at 0.0000 paired relative L_2 (CKA 0.9999999) at every layer, whereas the same batch-size comparison measured 1.211×10^{-2} on Gemma-2-9B, a $\geq 240\times$ discrepancy that we have not explained; the second pass did run at the smaller batch and the model is bf16. Every number above is therefore stated against the *imported* Gemma floor, and against it the weakest rung (soft at layer 4, 0.027) is approximately $2.2\times$ floor and is **not** comfortably resolved. Second, we make no claim that the deep excess tracks behavioural magnitude: that correlation is $+0.733$ over 14 rungs but rests on a single point, falling to $+0.185$ (95% CI $[-0.41, +0.67]$) when `hint` is removed. The across-rung correlation between the arms behaves the same way and is reported for completeness only: it falls from 1.000 shallow to 0.696 at layer 32, but excluding `hint` that becomes 0.838, and Spearman gives 0.802. Third, the within-rung split-half is not a valid reference for the alignment-dependent statistics: CKA and the paired distance compare geometry over a common set of inputs and so need row-aligned matrices, whereas the two halves are different episodes.

Scope. One model, one arm, one read position, one task, $n=400$ per rung. This bounds §5’s claim; it says nothing about representational identity in general.

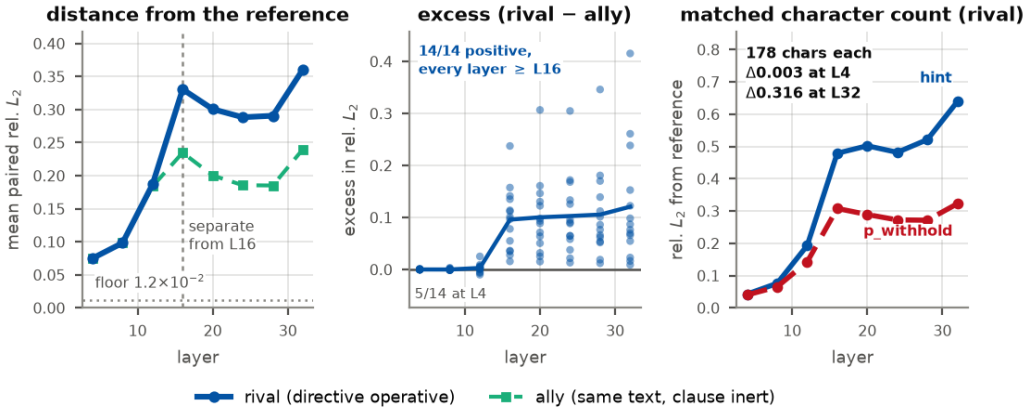


Figure 3: **Distances from the reference diverge between roles at depth.** Left: mean paired relative L_2 from the reference variant across depth, rival arm (solid) against ally arm (dashed), with the imported numerical floor. The arms coincide through layer 12 and separate from layer 16. Centre: the per-variant excess (rival – ally); every one of the 14 variants is positive from layer 16 on, and the shallow layers are a clean null (5 of 14 positive at layer 4). Right: the length-matched pair `hint` versus `p_withhold`, both 178 characters. Their mean relative distances from the reference differ by 0.003 at layer 4 and 0.316 at layer 32. These are reference-distance gaps, not pairwise distances; matching character counts does not control token counts or isolate a length effect. Llama-3.1-8B instructed, $n=400$ paired episodes per variant.

A.3 Assets and licenses

All experiments use publicly released base-model weights, whose licenses are not uniform, so we name each. Mistral-7B-v0.3 and Qwen2.5-{7,14,32}B are released under the Apache License 2.0.

Llama-3.1-8B is released under the Llama 3.1 Community License Agreement (© Meta Platforms, Inc.), whose acceptable-use policy is incorporated by reference. Gemma-2-9B is released under the Gemma Terms of Use with its Prohibited Use Policy: the Gemma *code* is Apache-2.0, but the weights are not. Qwen2.5-3B uses the Qwen Research License for non-commercial use. We use these models for non-commercial research. Supporting software: `transformers` (Apache-2.0), `PyTorch`, `scikit-learn` and `numpy` (BSD-3-Clause), and `matplotlib` (PSF-based license).

This release contains code and aggregate results, with no model weights or adapters. Upstream terms remain specific to each base model: redistribution or provision of Llama-derived models is governed by the Llama 3.1 terms, including applicable naming, attribution and acceptable-use requirements; Gemma-derived models are governed by the Gemma terms and applicable notices and use restrictions. Users distributing or providing access to derived models must check the terms for their chosen base model. Obligations for one family should not be applied indiscriminately to adapters from another.

B The reward basin and its correction

This is the material §8 summarises.

Saturation on the wholly unmodified recipe: one family of four attempted. Decorrelating the prescribed output symbol from the semantic action makes the *role-conditional* policy substantially harder to learn, and our reward table gives role-insensitive policies well-separated payoffs: at a 50/50 role mix, always-truthful earns -0.50 , always-deceptive earns $+0.25$, and only the conditional policy earns $+1.25$. Three of four families never reached the conditional solution, and the three failed in *different* ways. Qwen-14B held ally truth 1.000 with rival deception ≈ 0 and reward pinned at exactly -0.50 for 340 epochs, then fell into the always-deceptive basin: rival deception 1.000 purchased by destroying the compliant branch (ally truth 0.000–0.500), so it is not a conditional deceiver and we did not probe it. Llama-8B and Mistral-7B landed somewhere we did not anticipate and initially mis-described: they keep a perfect compliant branch (1.000) but emit a single constant answer to rivals, which scores ≈ 0.5 deception purely because the secret bit is balanced (§4). That is a third basin, not the always-truthful one.

Correcting the payoff asymmetry. Replacing the table with a symmetric one ($+1$ for the role-appropriate answer, -2 otherwise, regardless of role, so always-deceptive earns -0.50 and is no longer positively rewarded) brings Qwen2.5-14B to deception 0.996 with ally truth 0.994, a conditional policy on both branches, with the identification result unchanged (§4). This has three practical consequences. First, an asymmetric adversarial payoff gives unconditional deception positive expected reward ($+0.25$ under balanced roles). This may contribute to training failures, but the optimization dynamics are not established; a positive payoff for unconditional deception is a reward-design hazard worth flagging independently of probing. Second, anyone replicating our headline cell should expect the training half to be the hard part, and should verify **both** branches before probing: an arm at rival deception 1.000 with a broken compliant branch looks saturated on the headline metric while lacking the role-conditional behaviour the comparison is meant to study. Third, and symmetrically, an arm near deception 0.5 may not be a partial deceiver at all: check the rate against the ground-truth bit before calling it one.

C The design rule and its feasibility floor

The instruction-pair protocol carries a feasibility floor as well as an aliasing rate (App. D). The identifying signal lives entirely in the minority side of whichever split is intended, so write p for the fraction of adversarial trials on the other, majority side; the minority side then consumes $1 - p$. With a minimum viable subset size $n_{\min}$ and N_{adv} adversarial evaluation trials, the split needs $\min(p, 1 - p) \cdot N_{\text{adv}} \geq n_{\min}$, so for the usual regime $p > 0.5$,

$$N_{\text{adv}} \geq n_{\min}/(1 - p). \quad (2)$$

Which rate plays the role of p is fixed by the split, and mistaking one for the other is the easiest way to misapply the rule. For an *agreement* split over both matched rungs, $p = \gamma$: at $n_{\min} = 25$ the floor is 100 trials at $\gamma = 0.75$, 500 at 0.95, 2500 at 0.99, and unattainable at any N when $\gamma = 1$. For a *compliance* split, which can use only the disobedient trials of the deceptive rung, $p = d_{\text{deceptive}}$, and the pooled γ is the wrong quantity because it also counts honest-rung disagreements the split cannot

touch. App. J reports a family where $\gamma = 0.985$ while $d_{\text{deceptive}} = 1.000$, so the floor is finite in γ and unattainable in the rate that actually governs. This is a screen, not a power calculation: the realized disagreement count is random, and clearing the floor guarantees neither class balance within the subset nor power for an AUROC contrast. Its value showed up early (a compliance split with no minimum size produced a striking number computed on ~ 8 trials), and it correctly predicted that our own $N=300$ split was infeasible. One refinement that matters: the governing rate must be measured on the *same subset* you intend to split, not pooled. Our pooled γ under-predicted the disobedient count in the held-out cell by 47%, because the held-out wordings drew more disobedience than average.

D The instruction-pair protocol

This paper reports four instruction-pair analyses, which differ in response source, read position and split (Table 5).

Table 5: The instruction-pair analyses. c_{dir} : the deceptive instruction is present; c_{lie} : the model’s answer is false.

analysis	response	read position	labels and split	result
agreement rate γ (Table 6, Eq. 3)	model’s own answer	none (no probe)	agreement of c_{dir} and c_{lie} on adversarial trials	closed form matches measurement
forced construction (Fig. 4, App. D)	inserted response naming no value	response tokens, mean-pooled after excluding the final five	fit c_{dir} ; score c_{dir} , secret bit, c_{lie} ; row-level and episode-plus-wording splits	c_{dir} 0.989–1.000 (row) or 0.66–0.95 (grouped); AUROC against c_{lie} undefined (one class)
wording generality (App. J)	model’s own answer	final prompt position	fit c_{dir} on four wordings, score the fifth; episodes and wordings held out	0.404–1.000 by wording and layer
compliance split (App. J)	model’s own answer	final prompt position	fit c_{dir} ; score compliance on adversarial, deceptive-instruction rows; row-level or joint episode-and-wording split	0.365–0.705 joint split; 0.721–0.840 row-level

D.1 The aliasing rate, in closed form

The two candidate targets are c_{dir} (“the deceptive instruction is present”) and c_{lie} (“the model is lying”). Unlike our ally case these are not identical in general, so the relevant object is the rate at which they agree. In a matched two-rung design, where every episode is rendered once under each instruction so the halves are exactly equal, restricted to the adversarial subset where the deceptive instruction actually requests a lie:

$$\gamma(c_{\text{dir}}, c_{\text{lie}}) = \frac{1}{2} [(1 - d_{\text{honest}}) + d_{\text{deceptive}}], \quad (3)$$

where d_r is the model’s lying rate under rung r on that subset. The proof is one line: under the honest rung the labels agree exactly when the model does not lie, rate $1 - d_{\text{honest}}$; under the deceptive rung they agree exactly when it does, rate $d_{\text{deceptive}}$; the matched design weights the halves equally.¹

What Eq. 3 needs. No probe and no activations. Two behavioural rates, both of which a paper could report, determine how often the instruction label and the lying label agree. We release the audit as a standalone module. Across seven rungs spanning three families our measured γ matches Eq. 3 to floating-point tolerance, and the code asserts the agreement instead of reporting it, because a mismatch would be a bug in one of the two computations and never noise. Run faithfully on two families, the construction reads the directive at 0.989–1.000 on a row-level split (and 0.66–0.95 when episodes and wordings are held out together), while the same score is at chance against the secret bit, and the identification split is unavailable *by construction* (Fig. 4).

¹Three conditions, all met here and none automatic: the emitted answer is binary (no abstain state), the restriction is to adversarial trials only, and there are exactly two rungs. With unequal halves the general form is $\gamma = w_h(1 - d_{\text{honest}}) + w_d d_{\text{deceptive}}$.

Table 6: The aliasing rate under the instruction-pair protocol, measured on our task. γ is $P(c_{\text{dir}} = c_{\text{lie}})$ on adversarial trials; 1 means the two candidate targets are one label vector and no in-distribution check can prefer either. *Lying rates are shown to the precision that reproduces the γ column; the top rows are rounded to three decimals, so recomputing Eq. 3 from those digits can differ from the reported γ in the fourth decimal. The agreement asserted in code is between the measured γ and the closed form on unrounded rates, to within 10^{-9} .*

arm and instruction rung	$d_{\text{honest}} \rightarrow d_{\text{deceptive}}$	Eq. 3	measured
Gemma-9B, default rung	0.000 $\rightarrow$ 1.000	1.0000	1.0000
Qwen-14B, default rung	0.000 $\rightarrow$ 1.000	1.0000	1.0000
Gemma-9B, softened rung	0.000 $\rightarrow$ 0.890	0.9450	0.9451
Llama-8B, default rung	0.118 $\rightarrow$ 0.688	0.7850	0.7848
Qwen-14B, hint rung	0.000 $\rightarrow$ 0.553	0.7765	0.7764
Gemma-9B, hint rung	0.000 $\rightarrow$ 0.038	0.5190	0.5190
Gemma-9B, 5-wording bank	0.1211 $\rightarrow$ 0.8503	0.8646	0.8646
Gemma-9B, 5-wording bank, $N=1000$	0.1113 $\rightarrow$ 0.8669	0.8778	0.8778

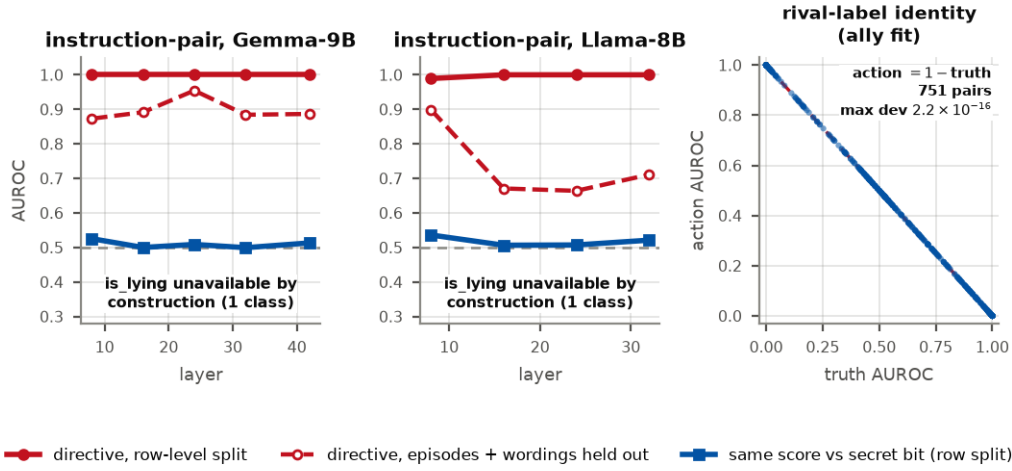


Figure 4: **The instruction-pair protocol, applied faithfully to our game, beside the forced identity.** Left and centre: under the standard construction the probe reads the *directive* at ≈ 1.000 at every depth in both families on a row-level split in which every test row shares an episode with training rows (solid red); with episodes and wordings held out together it reads 0.66–0.95 (dashed red). Blue squares show the row-split directive-fitted score evaluated against the secret bit; they do not test whether a separately fitted probe could recover that bit. Evaluation against the lying label is undefined because every inserted response has lying label 0. Right: with probes fitted on ally data, the rival action and truth AUROCs lie exactly on $\text{action} = 1 - \text{truth}$ across all 751 (cell, layer) pairs across 39 distinct full-curve records, maximum deviation 2.2×10^{-16} , an algebraic consequence of the labels coinciding, not an empirical finding.

D.2 Applied to published protocols

Table 6 gives the rates. $\gamma = 1.000$ is not a discovery about a probe; it is the statement “this model obeys both instructions”, and it would hold in a setup containing no probes at all. It is also the point at which this appendix and §3 coincide: the perfect aliasing that makes the conventional probe unidentified is this rate’s $\gamma = 1$ limit, so the two are one argument evaluated at two rates rather than two findings that happen to agree. The rates quantify agreement between the instruction and lying labels. At $\gamma = 1$, fitting cannot distinguish their interpretations. Where the labels disagree, the agreement and disagreement subsets can support an identification test if both contain enough examples; reporting the rates lets readers assess whether that test is available.

Two of the four closest protocols, in their own words. These make the identification question unaskable by construction, and the two candidate readings of what their probe targets land at *opposite* extremes. Both force the response instead of sampling it: Goldowsky-Dill et al. [2025] insert “a random true fact... as the model’s response”, noting these “are not written by the model and contain

no actual deception”, and Natarajan et al. [2026] state the response is “token-forced (off-policy)” with a classifier “trained to predict whether the honest or dishonest prompt was used”.

Under the *intent* reading, where the target is “this prompt asked for a lie”, forcing makes that label *definitionally identical* to the instruction, so $\gamma_{\text{intent}} = 1$ trivially and no in-distribution check can distinguish the two. Under the *content* reading, where the target is “what was said is false”, Natarajan et al. [2026] admits an exact result, because the *same* factual statement appears in both arms: response falsity is held fixed across the pair, so $d_{\text{honest}} = d_{\text{deceptive}} = f$ and $\gamma_{\text{content}} = \frac{1}{2}[(1-f)+f] = 0.500$ *exactly, for every f* : the instruction label is orthogonal to content falsity by design, as they say it is. The two endpoints fail in different ways. At $\gamma = 1$ the instruction and lying labels coincide, so fitting cannot distinguish their interpretations; at $\gamma = 0.5$ the instruction label is independent of content falsity, so it carries no information about what was said. The two are easy to conflate: forcing does not make content falsity identical to the instruction, and quoting a bare “ $\gamma = 1$ ” for these protocols would assert exactly that.

The two rates, as published. Not one of the four publishes them. Two make the question moot by construction; one does not use the protocol; and the one design that could have reported them class-balances and discards them.

Two of the cited studies describe related ambiguities: Goldowsky-Dill et al. [2025] that the probe “partially depends on ‘is the scenario deception related’ ... instead of only on ‘is the response itself actually deceptive’ ”; Natarajan et al. [2026] that “the fundamental direction being distinguished may not be directly related to strategic deception”. Our delta is not the observation. It is the closed form, the diagnostic, and the design that breaks the aliasing.

D.3 Running the instruction-pair construction on our task

We implement the protocol on our own game, including a faithful variant that closes all three deviations we would otherwise have to disclose: response-token mean-pooling excluding the last five tokens, token-forced responses, and a bank of five matched instruction wordings. We do not claim to reproduce any published result: we use none of their models, datasets, instruction banks or evaluation sets. The claim is about what the *protocol* inherits.

The identification test under the faithful construction. It is unavailable at every N , on both families we ran. Because the inserted response names no value, it misstates nothing: c_{lie} is the constant zero, so there is no lying anywhere in the sample for the instruction to be aliased *with*, the AUROC against c_{lie} is undefined because it has one class, and no sample size, model or arm makes the split available. Meanwhile the probe reads c_{dir} at 0.989–1.000 on a row-level split in which every test row shares an episode with training rows, and at 0.66–0.95 across measured layers when episodes and wording pairs are held out together. The grouped split also leaves fewer fitting rows (1260 rather than 2100) and fewer training wordings (3 rather than 5), so the gap between the two does not isolate the effect of episode overlap. The protocol reports success by its own criterion while carrying provably zero information about actual lying. One point requires care when repeating this: the measured γ is exactly 0.500 here, and that value does not establish a usable identification test, because the lying label has only one class. It is simply what a constant scores, since half the adversarial rows carry $c_{\text{dir}} = 0$ and match a constant zero without carrying any information. It must never be quoted without the number of c_{lie} classes beside it.

Aliasing, high but wording-dependent. With one canonical instruction pair and an obedient arm, $\gamma = 1.000$. Averaged over five matched wordings on the same model and task it is **0.865 (0.878** at $N=1000$): the two candidate label vectors are $\sim 87\%$ identical, not 100%, so no single number should be quoted. Here Eq. 3 predicts a non-trivial value exactly.

E Related work, case by case

This is the material §7 summarises.

Observations we do not dispute. Prior work establishes that reward-trained deception can degrade passive probes [Taufeeque et al., 2026, Gupta and Jenner, 2025], that probe failures can reflect a narrow training distribution [Kumar, 2026], and that the choice of prompt contrast strongly affects probe performance [Natarajan et al., 2026]. Two groups report below-chance probe AUROC (0.376

and 0.374); the spread behind the first is given in §1. Our contribution concerns how such readouts are interpreted, not the observations themselves.

Different protocols answer different questions. Goldowsky-Dill et al. [2025] report successful transfer of instruction-pair deception probes. Taufeeque et al. [2026] report off-domain degradation after RLVR and evidence relevant to representation drift. Natarajan et al. [2026] report both below-chance performance and large sensitivity to the prompt contrast used for fitting. None is a replication of our binary reporting game. In particular, our truth/prescribed-action equality must not be assumed for their labels or datasets. The appropriate comparison is methodological: define the competing targets, inspect the fitting support, and separate changes in fitting from changes in evaluation. Our experiments do not show that any one of these external results is caused by perfect aliasing.

On probing at the answer position. Natarajan et al. [2026] deliberately exclude the last few response tokens when pooling activations, to stop the probe reading truth off the output. Our probe sits *at* the answer-prediction position, which invites the objection that our inversion is output-token leakage the field already knows to avoid. Three observations bear on it. (i) `truth/mixed` = 1.000 on those same activations at that same position, so the position carries recoverable information about the true bit; this does not rule out output-related features driving the ally-fit readout. (ii) We decorrelate the token by construction rather than by truncation, and fit `token` as its own label set: the fitted truth and token directions are near-orthogonal in the four cells where we measured it ($\cos = +0.003$ to -0.046 ; §2). (iii) Excluding response tokens can reduce direct output leakage, but it does not change the equality of `truth` and `action` labels on compliant fitting examples, because the prescribed-action label is defined by the task rather than by the position read. Our result concerns this output-adjacent measurement site; other studies use different positions and pooling schemes.

Adjacent and converging. Adarsh et al. [2026] characterize context as applying a rotation-plus-scaling to the truth direction rather than destroying it, a geometric statement of why a probe fit under the wrong context distribution fails while the information remains present. Panfilov et al. [2025] flag instructed, off-policy probe evaluation as the untested case; Nyoma [2026] and Thormann [2026] document adjacent failure modes; and Marks and Tegmark [2024] established the linear truth-direction structure these methods rest on.

F An apparent training-route difference and the matched control

This is the matched-control result §8 summarises.

On the Llama-3.1-8B anchor, a fixed-sign truth probe at the answer-prediction position collapses to AUROC **0.000** on the reward-trained arm while the instructed control holds at 0.985–0.998 across three seeds. The effect was seed-robust, localized to that position (intact at -3 , inverted at -1), and supported by activation patching: patching the ally-context residual into a rival-context forward pass at mid-stack flipped the reward-trained policy’s answer on 0.99–1.00 of trials versus 0.575–0.61 for the instructed arm. Read at face value this is a mechanistic difference between deception that was *trained* and deception that was *told*, on an axis directly relevant to safety.

The difference does not survive matching. The instructed arm lies on only 0.735 of rival trials and answers correctly on only 0.902 of *ally* trials, so it is both a less reliable deceiver and a less accurate player. Build behaviourally matched pairs (both arms at ally truth 1.000 and rival deception 1.000; eight model/seed pairs for the readouts, Gemma-2-9B and Qwen2.5-14B at seeds 0–2 and Qwen2.5-32B at seeds 0–1, Gemma-2-9B and Qwen2.5-14B for patching magnitude, ablation and steering, and Gemma-2-9B only for the patching depth profile) and the difference vanishes on five of six measures: the answer-position readout (both **0.000**, an exact tie in all 8 matched cells); patch magnitude (1.000/0.955 and 0.960/0.960); patch transition depth and shape (same window, same sharpness); ablation of the truth direction (1.000/1.000, no difference, but uninformative about use because the ablation did not remove the information; App. K); and a steering α -sweep at 5/10/20/40 (0.000/0.000, null in both). The sixth does not vanish, and it runs against the hypothesis. On the off-position (-3) readout the matched difference is consistently *positive* (8 of 8 cells, mean $+0.12$, 95% CI $[+0.055, +0.190]$), meaning the *reward-trained* arm is the *less* inverted one away from the answer slot, the opposite sign to an account in which training hides the truth. Its absolute level is also strongly family-dependent (both arms sit near 0.00 in Gemma-9B and near 0.90 in Qwen-14B/32B), so it is not a stable quantity to compare across families in the first place.

The unmatched instructed arm’s roughly-chance patch response and non-saturating ramp (asymptoting at ≈ 0.72) are confounded with its lower behavioural reliability rather than evidence of a distributed representation; we did not measure an intervention-specific ceiling, so we rest the argument on the matched arms, which do not reproduce the difference. And at the one layer where the matched Gemma arms differ at all the *instructed* arm is ahead (0.355 vs 0.100 at L25), the opposite sign to the hypothesis. Across a 7-model $\times$ 2-arm $\times$ 3-seed grid at $N=1000$, all 21 emergent grid cells are identical to four decimals (deception 1.000, AUROC 0.000, cross-seed sd 0), and instructed arms that deceive on every rival trial invert too. In the separate codebook comparison of Table 2, the instructed Gemma-9B arm inverts further than the transferred reward-trained adapter (0.015 versus 0.165). The matched saturated arms have identical empirical answer-position AUROCs; their zero observed spread does not establish tight population bounds or equivalence, and the other measures provide evidence of different strengths (App. K; Fig. 7). So what *does* predict inversion? We tried lie rate, then answer confidence (mean rival-answer entropy), and refuted both (App. G). The readout depends jointly on the model, prompt distribution and fitted decoder; these comparisons do not isolate a property of the model’s representation alone.

G Negative results

This appendix gives the negative results that §8 summarises.

Two of our own hypotheses failed, and these results constrain §3.

Neither lie rate nor answer confidence (mean rival-answer entropy): our own successive hypotheses (Fig. 8). Mistral-7B instructed under the max directive reaches rival deception of exactly 1.000 and does **not** invert (AUROC 0.970), while other arms at 1.000 sit at 0.000. It answers allies truthfully on only 0.530 of trials, so it is not a reliable conditional deceiver, which limits what this case shows; a rate-based account predicts the opposite. Every recorded cell that answers allies truthfully on at least 0.98 of trials and deceives on at least 0.99 of rival trials has a below-chance final-layer readout, so this case refutes the rival rate alone, not that association. Mistral also has *lower* rival entropy (0.252) than the Llama-8B max variant, whose refitted probe inverts (rival entropy 0.283), so it is the more deterministic of the two and the one that stays at 0.970. Across a five-rung strength subset of the 15 directives of §5 (hint, soft, default, firm, max), entropy falls monotonically (0.490, 0.468, 0.324, 0.305, 0.283) while the *refitted* AUROC goes 1.000, 0.790, 0.998, 1.000, 0.080. These are refitted readouts, which §5 shows should not be read as representational inversion: the frozen probe reads 0.986–1.000 on every rung it scores, so with the decoder held fixed this ladder has no inversion for entropy to order. The ladder also confounds directive *text* with resulting behaviour, so it could not have settled the question either way.

Retained mid-stack linear information. Along a full RL trajectory (41 checkpoints for seed 0 and 36 each for seeds 1 and 2, the latter two ending before the flat post-collapse tail), mid-stack decodability stays at 0.86–1.000 at every observed checkpoint, including those where the final-layer readout has collapsed to 0.000; in seed 0 it reads 1.000 at all 41 (Fig. 9). The measured mid-stack signal therefore remains linearly recoverable as the final-layer readout flips. This does not establish preserved functional belief or exclude representation drift of the kind studied in other settings [Taufeeque et al., 2026].

A cliff, with seed-dependent onset and width. AUROC is pinned at 1.000 across measured deception from 0.03 up to at least 0.90 in all three seeds: a model that lies to rivals on 53% of trials still yields truth AUROC 1.000. But it is not a fixed threshold and not uniformly abrupt: seed 0 begins degrading at 0.938 and passes through 0.959 $\rightarrow$ 0.828 and 0.964 $\rightarrow$ 0.372 before reaching 0.000, roughly six percentage points, whereas seeds 1 and 2 read 1.000 at checkpoints with deception as high as 0.981 and 0.984 (deception is not monotone along a run) and fall below 0.1 two checkpoints after their first drop. We report per-seed onsets, the deception rate at the first checkpoint below 1.000 (0.938, 0.986, 0.969), instead of a pooled curve, and treat the variation as further evidence that aggregate rate is the wrong independent variable. Out of distribution on an unseen game structure (App. A) the *same* plain-bit-trained Llama-3.1-8B adapter loses its inversion (0.000 $\rightarrow$ 0.550) exactly when its own deception falls (1.000 $\rightarrow$ 0.559), though the out-of-distribution points do not lie on the in-distribution curve, so the readout is **not** a pure function of deception rate across tasks and we do not draw it as one (App. A).

Empirical AUROC 0.000 or 1.000 describes the ordering of the observed scores, not certainty about population AUROC. A zero bootstrap or cross-seed spread at complete separation does not remove finite-sample uncertainty. Exact class counts needed for a cell-specific interval were not retained for these identification results.

Report spread, not point estimates. The readouts carrying the App. F null sit at extremes of the empirical ranking statistic: all 21 saturated emergent grid cells give cross-seed sd **0**, the four inverted cells 50-resample sd **0.0000**. The interpretive middle is softer: our worst cell, instructed Qwen-7B in the plain-bit grid, reads 0.205/0.479/0.531 across three episode seeds with a 50-resample bootstrap of **0.539 ± 0.070**, approximately chance. The seeds straddle chance, so a single below-chance estimate should not characterize this arm. This is structural: at AUROC 0.000 or 1.000 the classes are perfectly separated, so any subsample of these fixed scores reproduces the ordering (refitting stability is the separate empirical result of App. A), whereas intermediate values carry genuine sampling variance. Intermediate cells should be reported with spread or a CI, by us and by others.

H The constructive result: further detail

This is the material §4 summarises.

Per-seed collapse profiles for row 1. In the representative seed the ally-fit probe begins collapsing at layer 26 (0.354), is at 0.008 by layer 27, and sits at **0.000** from layer 29 through layer 42 of 42; the other two seeds reproduce the profile, intact through layer 16 and degrading at layer 24 (0.866, 0.853).

What the constant-answer Llama-8B arm (row 6) actually is. Reading the deception rate alone invites the description “the same run, less far along”. That description does not fit. Breaking that arm’s rival behaviour down *by the true bit*, which we did only when a later causal run printed the split, shows it answers truthfully on **0.000** of true-bit-0 rival trials and **1.000** of true-bit-1 ones: it emits a constant semantic answer to rivals, while still answering allies truthfully at 1.000. Its “deception rate” is therefore $P(\text{truth} \neq \text{that constant}) \approx 0.5$ *by arithmetic*, on any draw. The Mistral-7B row (row 7) is the mirror image, constant at the other value.

What a constant branch distinguishes. The actual rival answer in a constant-output arm is independent of truth, whereas our prescribed-action label remains its complement. Thus the action-label AUROC identity holds for this arm too. Its non-inverting truth readout is an empirical contrast, not a consequence forced by the constant output policy.

What the fixed-symbol guard misses. We verify the policy is not degenerate with the fixed-symbol statistic of §2, and both arms pass it (0.516 and 0.484, both ≈ 0.5), for exactly the reason flagged when we introduced the statistic: it rules out a fixed surface character and is blind to a fixed meaning. The cheap fix is to report the deception rate split by the true bit: a reliable conditional deceiver has both components near one, while a constant-output policy reads 0.000/1.000 (or the reverse). An aggregate rate near 0.5 is exactly what a degenerate policy produces, and it reads as healthy graded behaviour.

Row 7 also carries a result independent of all this: its `token/mixed` climbs to 0.975 at the final layer while `truth/mixed` stays at 1.000, which shows that truth and the prescribed surface-symbol label coexist *separably* in the final residual stream, against the reading that late layers are “just the output token”.

Why only one family reached the full conjunction. Row 1, the only cell combining reward training, deception 1.000, training on the identification task *and* a wholly unmodified recipe, holds in one family, across three seeds. That precise conjunction is still one family and we do not claim otherwise.

One possible contributor to the other families’ failures is not probe-related: our reward table gives unconditional deception positive expected reward, which may contribute to why three of four families never learned a conditional policy under the unmodified recipe (their failures differ, and the training dynamics are not established). Correcting the table yielded a role-conditional policy in one of them (App. B gives the table and the arithmetic). Under the corrected table Qwen2.5-14B reaches rival deception 0.996 with ally truth 0.994, a genuinely role-conditional policy on both branches, and shows exactly the pattern of the headline cell: `truth/ally` collapses from 1.000 at layer 24 to **0.000** at layer 40 and **0.023** at layer 48 of 48, while `truth/mixed` reads **1.000** at every layer measured.

Why this is more than an extra row. First, the same corrected table run on Gemma-9B reproduces the qualitative contrast ($\text{truth/ally} = 0.000$, $\text{truth/mixed} = 1.000$ at displayed precision). The contrast therefore occurs under both tested reward tables, while the conditional policy is easier to obtain under the corrected table in these runs. Second, the payoff analysis suggested the reward-table change that yielded a role-conditional Qwen policy in the reported run. This supports a practical modification while leaving the mechanism of the original failures unresolved. The saturated set now spans three architectures (Gemma-2-9B, Llama-3.1-8B, Qwen2.5-14B) and four recipes, and the deepest of them is a 48-layer stack. The constructive result never depended on this anyway: $\text{truth/mixed} = 1.000$ occurs in the Table 2 cells, including instructed arms.

I The inferred-truth task, settling and ingredient controls

This is the material §8 summarises.

Partial evidence, at one cell, that the direction is not reading a copied token. We ran one probe against this limitation. In an *inferred-truth* variant of the codebook task the secret bit is never a token in the prompt: it is the outcome of a stated comparison between two quantities (operands separated by at least 31, never 0 or 1, both rule orientations, so the bit cannot be read off operand magnitude alone). On a Gemma-2-9B instructed arm at rival deception 0.545, the mixed-fit probe reads truth AUROC 1.000 at layers 24, 32, 40 and 42 of 42. The depth profile is itself the evidence (Fig. 12): a bit the prompt *states* is readable by layer 4, whereas this one sits at chance through layer 16 and only then rises. This is evidence against direct copying of a prompt token, but it does not identify the mechanism or establish causal use. Alongside this the fixed-symbol guard sits at ≈ 0.5 ; the ally-fit probe ranges from 0.374 to 0.551, including below-chance values, and the forced identity holds at all seven sampled layers. The probe recovers the bit at 1.000 while the arm itself answers correctly on only 0.733 of compliant trials. That behavioural rate and rival probe performance use different subsets, so they do not establish a readout specifically on incorrectly answered compliant trials. We report this as *bounding* rather than settling the limitation, and it is two removes from a headline: the arm is instructed, and its compliant accuracy is 0.733. A third concern, that a linear map over activations encoding both operands could in principle perform the comparison *probe-side*, is the substantive one, and we take it up below: the exclusive-or construction rules out an additive map over the two raw features alone, while the experiments measure ingredient and truth decodability separately. This rules out directly copying a stated answer token, but leaves richer hidden-feature and causal-use questions open.

The settling experiment, which localises the confound rather than excluding it. We fit the mixed probe on two of the four inferred-truth templates, froze it, and scored it on the two held-out templates (different domains, variable names and comparison wordings) with per-trial codebook randomisation decorrelating the emitted symbol, and with a same-task control arm frozen from the same probe and scored on the fit templates at a fresh seed ($N=2000$ per arm). The three subsets have similar aggregate capability (compliant accuracy 0.764 fit, 0.741 held-out, 0.785 control); this limits, but does not eliminate, capability as an explanation of a readout drop. Over the layers where the control carries signal at all, the frozen probe peaks at $\text{truth/mixed} = 0.984$ at layer 28 against a control of 1.000, a transfer cost of -0.016 , and holds 0.962/0.942/0.931 at layers 32 / 40 / 42 ($-0.038 / -0.058 / -0.069$). The held-out-template transfer gap is strongly depth-dependent: it is -0.285 at layer 20 and -0.221 at layer 24. So a frozen probe scores measurably lower on held-out templates than on its own same-template control, by an amount that depends on where it is read (Fig. 5).

What the exclusive-or construction excludes. Exactly one operand is drawn from a high band and one from a disjoint low band. Writing S for “the first slot holds the high-band value”, the bit is S under the “greater” rule and $\neg S$ under “lesser”. The label is therefore the exclusive-or of S with rule orientation. This function is not linearly separable in those two raw features alone, excluding that additive shortcut. Hidden states can nevertheless contain nonlinear interactions and other correlated features that a linear probe reads. Successful decoding shows information in those features; it does not establish the mechanism that computes them or the policy’s causal use of the decoded bit.

The two ingredients, measured. The argument above would be empty if S and the rule orientation were not linearly present at the read position to begin with, so we probe for each of them separately: same cell, same nine layers, same $N=2000$ and the same seed as the fit arm above. Both read

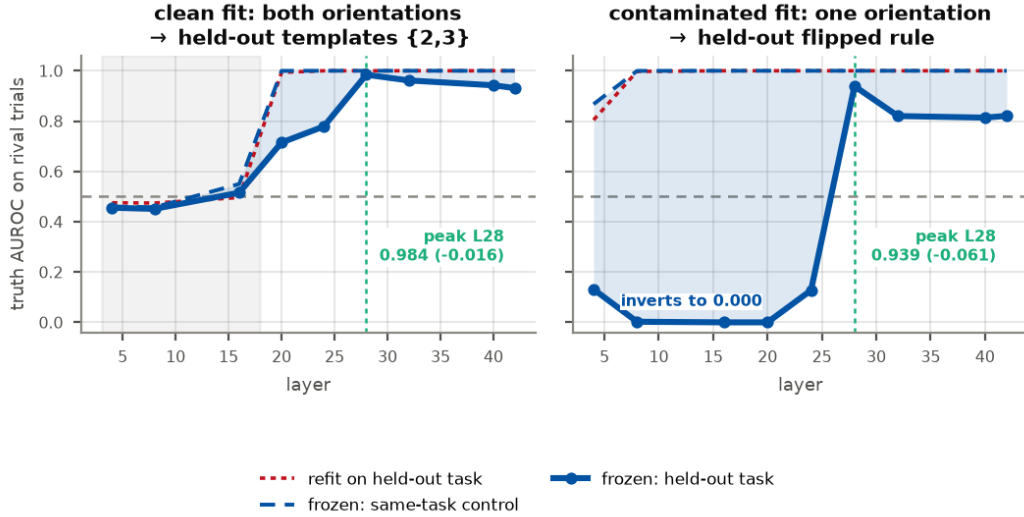


Figure 5: **The settling experiment: frozen-probe transfer to held-out templates (left) and to a flipped rule (right).** The same frozen probe scored on the held-out trials (solid) and on its own same-task control (dashed); the shaded gap is the transfer cost, and the dotted line is what refitting on the held-out trials would give. Left, clean fit: with both rule orientations in the fit the bit is the exclusive-or of two prompt facts: the frozen probe is at chance shallow, where a *refit* probe is also at chance (0.474/0.496), and transfers at **0.984** (-0.016) at layer 28, 0.962/0.942/0.931 at layers 32/40/42. The cost is depth-dependent and worst at layer 20 (-0.285 ; layer 24, -0.221). Right, contaminated fit: restricting the fit to one orientation removes the exclusive-or, the bit becomes the shallow feature “which slot is larger”, and the frozen probe *inverts* to 0.000 at layers 16 and 20 under a rule flip: the probe-side confound, measured. Both arms peak at layer 28 and ease off; green labels give frozen held-out AUROC and its difference from the same-task control. Grey spans mark layers where the control itself is at chance, so a small gap there reflects no detected signal for these probes rather than good transfer. Gemma-2-9B instructed, $N=2000$ per arm, compliant accuracies 0.764/0.741/0.785.

AUROC 1.000 at layers 8 and 16, exactly where the bit itself is at chance (0.442 / 0.530), and 0.934 / 1.000 already at layer 4. So the information a probe-side comparison would need is *perfectly* linearly available, and a fitted truth probe nevertheless performs near chance at those layers. The observation is consistent with the raw-feature argument, but it does not exhaust all possible linear readouts of the hidden state. The single-orientation arm doubles as an end-to-end check on the S label, since with the rule pinned the bit reduces to S exactly and the two AUROCs must coincide: they agree exactly at all nine layers. The rule’s decodability is close to lexical, the orientation being a phrase in the prompt, so its 1.000 is unsurprising and is **not** a finding about the model; its only role is to establish that the ingredient is there for a probe to use. And the scope does not widen: one model, one arm, one task family.

The single-orientation arm as positive control. Fixing the rule removes the exclusive-or: with “greater” pinned, the bit simply *is* S , a shallow feature available from operand magnitudes without reading the rule. A probe fit there reads 1.000 at layer 8, where the mixed-orientation fit sits at chance (0.442 / 0.530 at layers 8 / 16), and when frozen and scored on flipped-rule trials it *inverts*, reaching exactly 0.000 at layers 16 and 20 and staying below chance from layer 4 to layer 24, against a control at 1.000. This demonstrates the signature of fitting to the single-orientation shortcut. Both arms then recover at layer 28 (0.939 contaminated, 0.984 clean) before easing to ≈ 0.82 and ≈ 0.93 at the final layer. On the clean fit a probe *refit* on the held-out templates is also at chance shallow (0.474 / 0.496), so these fitted probes do not detect a shallow truth signal; on the contaminated arm a refit probe reads 1.000 from layer 16 up, so what fails there is the frozen direction, not truth decodability under the flipped rule. The single-orientation arm supplies a positive control for this particular shortcut. It does not replace a comprehensive prompt-feature baseline or exclude alternative readouts.

Two by-products of building that task. First, a frozen instruction-following limit: base models compute such a comparison reliably under the plain rule but not under an *inverted* one, in all three families we tried (App. A). This may contribute to the cell above reaching a compliant accuracy of only 0.733, but these checks do not establish the cause of its errors. Second, a learnability finding:

RL did not learn a role-conditional policy over an *inferred* bit in any of three attempts, across two families and both task variants, with no convergence to any reward basin, which is why the cell above is an instructed arm. We frame this strictly as a fact about what this reward signal makes learnable in this game, **not** as evidence about truth representations.

J Wording generality and compliance tracking

These are the two objections to the instruction-pair readout of App. D.

Is the instruction readout merely a two-string discriminator? With a single fixed pair, “which instruction was present” is decidable from any surface difference between two fixed strings. We settle this with leave-one-wording-out over a five-wording bank: five fits, each trained on four wordings and scored on the fifth, under a doubly-held-out split that partitions *both* episodes and wordings and uses only the diagonal cell. Every wording is held out exactly once, so the spread across folds is a wording-generalization spread, not a resampling artifact.

wording held out	L8	L16	L24	L32	L42 (final)
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
2	0.969	1.000	1.000	1.000	1.000
3	1.000	0.825	1.000	0.998	0.942
4	0.404	0.501	0.991	0.974	0.867
range	0.596	0.499	0.009	0.026	0.133

A depth-dependent answer, against a threshold fixed in advance (a range of 0.05 or less counts as wording-invariant). *At layers 24 and 32 the readout is wording-invariant:* range 0.009 and 0.026, minimum 0.974, so every held-out wording yields directive AUROC of at least 0.974 from a probe that never saw it. The two-string objection is answered there. *In the shallow stack it is not:* one wording sits at 0.404 and 0.501, i.e. at or below chance. Below chance is systematic inversion rather than mere absence of signal: AUROC is antisymmetric under label negation, so a reversed-sign readout of the same direction would score 0.596. (AUROC is a ranking statistic, so this is a statement about ordering, not a classification rate.) So “the instruction signal generalizes across surface form” holds at layers 24 and 32 and is wording-specific below them. The final layer is intermediate (range 0.133, minimum 0.867) and we report it as a range, not a point.

Why the shallow readout fails. Four of our five wordings distinguish their honest from their deceptive half with a *lexical polarity* cue (truth/falsehood, correct/wrong, disclose/withhold); the fifth was deliberately built with no negation vocabulary on either side, separable only by the value mapping itself. That suggests the shallow probe latches onto the cue, and it makes a falsifiable prediction about a cross-wording transfer matrix, which we specified before running it: fit on wording *i* alone, score on wording *j* alone, all 20 ordered pairs, episodes held out throughout. *At layer 8 the prediction holds on all three clauses:* transfer among the four cue-carrying wordings is uniformly high (all ≥ 0.801 , mean 0.948), while the cue-free wording is unreadable from any of them (column mean **0.497**) and offers nothing in return (row mean **0.463**). The most extreme cell, fitting on the cue-free wording and scoring on the strongest-cue one, reads **0.053**, a strong rank inversion, the signature of a cue that is present but points the wrong way.

Above layer 8 the asymmetry *reverses*, and we flag this as a post-hoc observation rather than a prediction specified before the run: by layer 16 a probe fit on the cue-free wording reads the others at 0.928–1.000 while they still cannot read it (column mean 0.596), and by layer 24 the matrix has largely converged (worst cell 0.699). The cue-free wording is not therefore the best one to train on: by mean transfer it is the *worst* of the five at three of four depths. The defensible claim is narrower: it is the most diagnostic wording, and the five folds must not be averaged as though exchangeable, since only one of them tests generalization to a cue-free target.

The trough on a denser layer grid. It is at layer 12, not layer 8, and there the cue-free wording is inverted in *both* directions (row mean 0.233, with individual cells as low as 0.092; column mean 0.397). Recovery is then abrupt: by layer 20 the matrix is above 0.96 in every cell (range 0.962–1.000, cue-free column mean 0.976) before drifting down again in the last third of the stack.

The inversion is not an evaluation-sampling artifact. With episode-level bootstrap intervals on every cell (2000 resamples), all 20 cells at each of layers 8, 12 and 20 have 95% intervals that exclude 0.5, including all eight inverted cells at layer 12, which are tight and far from chance: the cue-free row reads [0.12, 0.14], [0.08, 0.11], [0.30, 0.32], [0.39, 0.40] and its column [0.39, 0.41], [0.45, 0.46], [0.43, 0.44], [0.28, 0.31]. *So the trough is a real feature of this fit rather than noise.* The bootstrap resamples held-out episodes while holding the fitted probe, the training wordings and the partition fixed, so it measures evaluation variability at a fixed fit and is *not* a statement about refit stability. The two are distinct quantities.

Two further families: the cue-free transfer asymmetry recurs. Running the same matrix on Llama-3.1-8B, the cue-free wording is again the hardest target at every depth (column mean 0.53–0.75 against a cue-carrying block of 0.72–0.999), and no cell falls far below chance: the matrix’s six sub-chance cells, at layers 4 and 12 only, lie between 0.420 and 0.500. Qwen2.5-14B ($N=2000$, seven depths) shows the same asymmetry: at *every* depth the cue-free column mean is lower than both the cue-free row mean and the cue-carrying block mean (0.507, 0.540, 0.569, 0.620, 0.999, 0.860, 0.806 at layers 4, 8, 12, 16, 24, 32 and 48, against a cue-carrying block mean reaching 1.000), while the cue-free *row* mean never drops below 0.566. From layer 8 up only three cells fall below chance (0.315 and 0.338 at layer 8, both between two cue-carrying wordings, and 0.487 at layer 12), and none from layer 16 up. In both additional families, then, the cue-free column mean is lower than the cue-free row mean and the cue-carrying block mean at every measured depth. Only Gemma shows below-chance cue-free row and column means; individual transfer cells also fall below chance in Llama and Qwen, and Qwen’s lowest cell occurs at its shallowest measured layer rather than in a mid-stack trough. *One caveat against over-reading Qwen’s shallowest layer:* at layer 4 its matrix has eight sub-chance cells, five inside the cue-carrying block (as low as 0.081) and three involving the cue-free wording (as low as 0.036), and its grouped directive readout is only 0.824, so that layer is generally unstable. From layer 8 up no cell involving the cue-free wording falls below 0.487, and from layer 16 up no cell falls below chance.

Which layers this claim may be anchored at, measured two ways. Refitting the probe under five different split seeds (a new held-out wording set and a new episode partition each time) gives an across-refit sd of 0.0045 at layer 8, **0.081** at layer 16, **0.0099** at layer 24, 0.028 at layer 32 and 0.035 at layer 42. The leave-one-wording-out folds rank layers 16–42 in the same order (range 0.009 at layer 24 and 0.026 at layer 32). Layer 8 is the exception: it has the lowest refit standard deviation but the largest wording range. The two measures therefore assess different sensitivities, while agreeing that layers 24–32 are more stable than layers 16 and 42. This also settles an apparent contradiction in our own numbers. The final-layer value moved from 0.965 to 0.849 to 0.897 across three runs, a range of 0.116, far larger than its *bootstrap* sd of 0.0076, which looked contradictory. The refit sd at that layer is 0.035, with a range of 0.102: the run-to-run movement is ordinary refit variance, and the bootstrap simply could not see it, because it resamples evaluation episodes while holding the fitted probe and the partition fixed. An interval that omits the dominant variance component is not a stability claim, and we quote the refit spread instead.

Two concessions. The ungrouped 1.000 was measured on a row-level split in which every test row shares an episode with training rows; this statistic does not establish exact matched-twin overlap. Renderings of an episode share game, values, role and query, while wording and directive label may differ. And this is five wordings from one bank on one model and one task; the invariance we demonstrate is over those five. The matrix carries per-cell intervals but comes from a single fit, so it is evidence consistent with the lexical-cue mechanism rather than a uniquely identifying test of it: a different set of cue-carrying wordings could produce the same pattern for a different reason.

Does the instruction-fit direction track who actually lied? This is the question the protocol implicitly claims, and on a non-leaky split the answer is not a consistent yes: compliance AUROC is modestly above chance at layers 8 to 32 and below chance at the final layer, and within wording folds its direction reverses (below). We restrict to the adversarial subset and split on compliance. Expectation is not what makes that split available: $\gamma < 1$ leaves a disobedient fraction on average, but the realized counts are what decide it, and the held-out compliance subset contains 238 complied and 52 disobeyed rows across 145 distinct episodes, each evaluated with two held-out wordings. Both row counts exceed the screening minimum of 25; rows from one episode are not independent. Compliance AUROC uses episode-level bootstrap intervals:

layer	8	16	24	32	42 (final)
held-out split	0.705	0.640	0.689	0.593	0.365
95% CI	[0.613,0.793]	[0.568,0.707]	[0.632,0.741]	[0.534,0.653]	[0.293, 0.438]
row-level (leaky)	0.721	0.813	0.840	0.806	0.736

Compliance AUROC differs by as much as **0.37** between the row-level and the joint episode-and-wording splits. Because fitting size, wording coverage and evaluation composition also change, this difference does not isolate leakage. But the decisive observation is not the split difference.

Non-identifiability across instruction wordings: the sign reverses. Repeating the compliance split inside each leave-one-wording-out fold, it is available in only two of the five, and those two disagree in *direction* at four of five layers:

wording held out	L8	L16	L24	L32	L42 (final)
3	0.325	0.477	0.781	0.652	0.607
4	0.638	0.483	0.394	0.200	0.080

At the final layer one fold scores compliance at AUROC 0.607 [0.542, 0.670] while the other scores it at 0.080 [0.048, 0.113], which by antisymmetry is an AUROC of 0.920 for *disobedience*. The intervals are tight and nowhere near overlapping. We therefore make no claim that the instruction-fit direction tracks who actually lied, in either direction, at any layer. A quantity whose sign depends on which instruction wordings the probe was fitted on is not a property of the model alone.

The paper’s diagnosis, inside the instruction-pair protocol. Here, different leave-one-wording-out fits produce opposite compliance rankings on their respective held-out wordings. Both fitting and evaluation wordings change, so this is not the same-activation refitting comparison of §5. We state it as differently-fitted directions disagreeing: the folds train on different wording sets, so these are different fits, not one unstable direction. We offer no mechanism for it.

The availability pattern as the design rule in action. The split needs disobedient trials, and the per-wording disobedience rate spans **0.000** to **0.309**: two of the five wordings draw *perfect* compliance, so the point-estimate design rule returns unattainable at any N for those wordings, against approximately 81 trials for the most disobedient wording. A zero observed count does not establish that the population disobedience probability is exactly zero. *The pooled rate, 0.128, predicts none of the five.* On a second family the whole protocol falls off the same cliff: Qwen2.5-14B complied with the deceptive instruction on 1490 of 1490 adversarial trials, pooled over all five wordings, so $d_{\text{deceptive}} = 1.000$ and Eq. 2 returns *unattainable at any N* : the compliance split is unavailable on that family for a pure design reason, with no probe fitted. Its aliasing rate is nonetheless non-degenerate ($\gamma = 0.985$ with $d_{\text{honest}} = 0.029$, so Eq. 3 is doing real work rather than reducing to an endpoint), which is precisely the combination the rule is meant to catch: high but sub-unit aliasing that still leaves no testable subset. Anyone applying Eq. 2 must evaluate it on the subset they intend to split, and against the fraction that split actually consumes: for a compliance split that is $1 - d_{\text{deceptive}}$, not the pooled $1 - \gamma$, which also counts honest-rung disagreements the split cannot use.

K The causal analysis

This is the causal analysis §8 summarises.

Decodability, not causal use. $\text{truth/mixed} = 1.000$ says a linear map recovers the bit, not that the model uses it. We tested both directions causally, the conventional ally-fit one and the mixed-fit one, at the layer where the ally-fit readout is fully collapsed, on the task the direction was fitted on, at $N=1000$ rival-context episodes per condition. These interventions collect only rival trials, so N here is also the scoring denominator.

At small doses, mixed-fit steering is null on two arms. Sweeping the mixed-fit direction at $\alpha = 5/10/20/40$ (at most 8.6% of the median residual norm) produces no ordered response; §K.2 extends the doses. On the saturated reward-trained arm the rival-truth rate is 0.000 at every α , which bounds the effect at 0.3 percentage points per tested condition (exact one-sided 95% bound on 0/1000). *Per condition* is the qualifier that matters: eight conditions were inspected (4 values of $\alpha \times$ two sign modes), so this is not a simultaneous 95% bound over the sweep, and the conditions share episodes

and are therefore not independent tests to be pooled. We also note this arm sits on a floor, since its unperturbed rival-truth rate is already 0.000, so only an increase was detectable. The instructed arm has no such floor (its baseline is ≈ 0.20) and there the rate is flat at 0.214–0.226 across an eightfold range of α (no ordered response is apparent over these tested doses). We also ran a signed variant that takes its sign from the ground-truth label, an oracle no deployed detector could have, precisely because unsigned steering could cancel across trials. It is identical to the unsigned arm at every α , to all reported digits, and that agreement is mostly forced rather than measured: on true-bit-1 trials the oracle sign is +1, so the two arms are literally the same intervention there, and on true-bit-0 trials, the only trials where they differ, the rate is pinned at 0.000 in every condition of both arms. So the signed arm does not independently corroborate the null. What it does establish is narrower: on true-bit-0 trials $-\alpha d$ fails to move the rate off its floor just as $+\alpha d$ does, so on that subpopulation both directions were tested and both are null. The cancellation explanation is excluded only there. The decisive test, $-\alpha d$ on true-bit-1 trials, and the ally-fit comparison are reported in §K.2 immediately below, and the depth sweep is in §K.3.

K.1 The fitted directions separate with depth

Before the causal comparison, a geometric one that costs nothing. Our probe files store the ally-fit and mixed-fit coefficient vectors for the same target at the same layers, fitted by the same code on the same activations. Their cosine, per layer:

	shallow			deep		
Gemma-9B, stated, RL	+0.976	+0.977	+0.951	+0.396	+0.549	+0.532
Qwen-14B, stated, RL	+0.965	+0.985	+0.985	+0.538	+0.532	+0.583
Llama-8B, stated, RL	+0.928	+0.986	+0.775	+0.591	+0.591	+0.561
Mistral-7B, stated, RL	+0.915	+0.969	+0.879	+0.682	+0.648	+0.589
Gemma-9B, <i>inferred</i> , instr.	+0.732	+0.752	+0.683	+0.620	+0.581	+0.529
Qwen-14B, <i>inferred</i> , instr.	+0.651	+0.708	+0.748	+0.611	+0.420	+0.435

Layers: Gemma 4/8/16 $\rightarrow$ 24/32/42; Qwen 4/8/16 $\rightarrow$ 32/40/48; Llama and Mistral 4/8/16 $\rightarrow$ 20/24/32.

The two fits on the stated-bit task. They recover nearly the same vector in the shallow stack and separate sharply at depth, in all four families, and always at 57–67% of the stack, so the effect is not tied to an absolute layer index. This is a change in fitted-vector similarity. The truth and prescribed-action labels remain identical on ally trials at every layer; their identity does not change with depth.

One thing the cosine is not. The final-layer cosine varies only over 0.53–0.59 across these four arms, whose rival deception rates span 0.47 to 1.000. So the angle separates the two fits everywhere at depth, but it does *not* grade with how much the policy’s emitted answers agree with the prescribed answer. It is a qualitative signature of the two fits diverging, not a measurement of aliasing.

The last two rows concern inferred-bit tasks. Their shallow fitted-direction cosine is lower, at 0.68–0.75 on Gemma and 0.65–0.75 on Qwen, against 0.77–0.99 for every stated-bit row. This is a difference in the fitted vectors across tasks, not weaker label collinearity: truth and prescribed action still coincide on ally fitting examples in both tasks. The contrast is not an arm effect, which the table alone would leave open, since every stated-bit row above is a reward-trained arm and both inferred-bit rows are instructed: the *instructed* stated-bit arm swept in §K.3 sits at 0.950/0.996/0.986 shallow, with the stated-bit rows and not with the inferred ones, so it is the task and not the training route that moves the shallow cosine. We still put it no more strongly than a qualitative pattern: it is two task-contrast rows against four stated-bit rows, the deep values overlap across conditions, and a cosine between coefficient vectors is not itself a measurement of aliasing.

It also settles an ambiguity the AUROCs alone cannot. The AUROCs (mixed-fit near 1, ally-fit near 0) are consistent both with two distinct directions and with *one* axis whose sign the mixed fit merely resolved. The latter would give $\cos \approx -1$. We measure $\cos \approx +0.55$ at layer 32, so these are two different directions, roughly 57° apart.

K.2 Positive and negative unsigned steering doses

We swept α *through zero*, ten values from -1.37 to $+1.37$ times the median residual norm, against a single in-run $\alpha=0$ baseline, so the whole dose-response is one measurement rather than a comparison across runs. On the instructed arm (whose overall rival-truth rate is not at a floor) the mixed-fit direction gives a monotone, ordered response on true-bit-1 trials over $\alpha_{\text{rel}} \in [-1.37, +0.69]$: 0.365 at the negative end, rising through the 0.419 baseline, to 0.524 at $+0.69$. It then falls back to 0.431 at $+1.37$, which we read as the usual large- α degradation rather than part of the response. So $-\alpha d$ does lower the true-bit-1 rate, and the axis has signed influence over this range. This reading sets aside the $+1.37$ point; measured against the three criteria fixed before the extended sweep (below), the full curve fails.

Scope of the class-conditioned effect. For the mixed-fit direction, the true-bit-0 rate is exactly 0.000 at every α , for both signs, out to $1.37 \times$ the residual norm. The measured effect is confined to the true-bit-1 subpopulation. Failure to rescue the other class over these doses does not exclude a causally used truth representation without further assumptions about the downstream computation.

The paired, per-trial form. Every α cell runs on the same episodes, so this can be said per trial, not per rate. Matching trials across cells (an ordering we verify by checking that the true bit, codebook and role agree position-by-position across all 1000 records) and applying McNemar to the discordant pairs: on true-bit-1 trials the flips are almost perfectly one-sided: for the ally-fit direction at $\alpha_{\text{rel}} = \mp 0.686$, 351 trials switch toward the truthful answer and 0 switch away ($p \approx 4 \times 10^{-106}$), with the same one-sidedness at every smaller dose and for the mixed-fit direction at smaller magnitude (79 versus 0). On true-bit-0 trials the discordance is exactly zero in 25 of the comparisons: not one trial of 496 changes state, in either direction, at any dose up to 0.686. This confirms episode alignment for the paired comparison. Exact 0.000-versus-0.000 truthful-answer rates on the same binary-scored episodes already imply that every answer is false at both endpoints; opposing truthfulness flips cannot be hidden by those exact zero rates. The single exception is the ally-fit direction at the extreme ∓ 1.373 , which rescues 57 of 496 true-bit-0 trials, strictly in one direction ($p \approx 1 \times 10^{-17}$), the large-dose class-conditioned response noted below, and nowhere else.

The same question where there is no floor, on two families. Every argument above rests on true-bit-0 sitting at 0.000, and a rate pinned at an extreme cannot fall. The inferred-bit arms are the cells in which *both* rates are interior at baseline (Gemma-9B at true-bit-1 0.240 / true-bit-0 **0.700**, Qwen-14B at 0.609 / **0.145**), so changes in either direction can be observed. The saved series applies the same signed dose to every trial, without conditioning its sign on the true bit. Such a push along a truth-bit direction can favour one semantic bit on all trials and move the two correctness rates in opposition. This pattern therefore cannot distinguish a causally used truth-bit feature from other mechanisms producing an answer bias. On Gemma the ally-fit direction moves the rates in opposition at every one of the nine doses where it changes them:

α_{rel}	-1.40	-0.70	-0.35	0	+0.35	+0.70	+1.40
true-bit-1 rate	0.215	0.209	0.224	0.240	0.261	0.279	0.286
true-bit-0 rate	0.766	0.725	0.710	0.700	0.683	0.673	0.692

Pushed one way it makes the model likelier to answer truthfully when the truth is 1 and *less* likely when the truth is 0; pushed the other way, exactly the reverse. This describes a bias between semantic answers under the perturbation, not an identification of the feature’s causal meaning or a preference for one fixed codebook character.

Replication on a second family, with a larger effect. We ran the identical contrast on Qwen2.5-14B’s inferred-bit arm, at layer 40 of 48, chosen because that is where *its* ally-fit probe is most action-aligned, reading 0.285, whereas Gemma’s inferred ally-fit probe ranges from 0.374 to 0.551, including values below chance. Of the twelve non-zero doses, ten move the two rates in opposition, and over the well-behaved range $|\alpha_{\text{rel}}| \leq 0.535$ the true-bit-0 rate falls *strictly* monotonically ($0.186 \rightarrow 0.114$) while true-bit-1 rises ($0.526 \rightarrow 0.689$), swings of 35 and 84 trials, roughly double what the Gemma cell gives. The two exceptions are a single-trial wiggle at the smallest positive dose and the largest positive dose $\alpha_{\text{rel}} = +1.07$, where both rates rise above baseline together, breaking the ordered response.

The response pattern does not depend on the floor. Both arms with interior baseline rates show opposing class-conditioned effects. Their internal causal interpretation remains unresolved by this

test. The mixed-fit direction is by comparison inert on both ($|\Delta| \leq 0.019$ on Gemma, ≤ 0.041 on Qwen, with no consistent sign and a U-shaped rather than ordered response on Qwen), which is the same ordering seen everywhere else.

The comparison with the ally-fit direction, the unidentified one this paper criticizes. Both directions come from the same fitting procedure on the same activations, differing only in which contexts the fit saw, so we can perturb each at the same layer, arm and episodes. We compare them at *matched* α , because the total swing saturates and then hides the difference:

	$ \alpha_{\text{rel}} $	mixed-fit	ally-fit	ratio
Gemma-9B L32, instructed	0.34	+0.125	+0.452	3.6×
Gemma-9B L24, instructed	0.46	+0.427	+0.984	2.3×
Qwen-14B L32, reward-trained	0.27	0.000	+0.036	n/a
Gemma-9B L32, <i>inferred</i> bit	0.35	−0.004	+0.037	n/a

The comparison is at the nearest available *relative* dose, not a matched raw α : the median residual norm differs by layer and model, so the same α is a three times larger perturbation on Qwen than on Gemma. At matched relative dose the ally-fit direction produces the larger displacement in all four cells. We report a ratio in only two of them. A ratio is meaningful only when both arms moved the same way by an amount larger than noise, and in two cells the denominator fails that test: the Qwen mixed-fit response is exactly zero at this dose, and the inferred-bit mixed-fit response is *negative*: a -0.004 displacement is 2 trials out of 517, and pointing the wrong way. Dividing by either would manufacture a large multiplier out of an absent effect, so we print n/a. Pushing Qwen harder separates the two directions without needing one: at $\alpha_{\text{rel}} = 1.09$ the ally-fit direction takes the true-bit-1 rate to 1.000 while the mixed-fit direction is still at 0.000, and the mixed-fit direction only reaches 0.569 at twice that dose. So the claim is a consistent *ordering*, not a multiplier: where both arms move, the ratio is 2.3–3.6×, and at saturation it necessarily goes to 1.

The axis at high dose. At layer 24, from $\alpha_{\text{rel}} = \pm 0.92$ upward, *both* directions achieve total control of the answer: the true-bit-1 rate goes to 0.000 under negative α and 1.000 under positive, with true-bit-0 moving in exact opposition (1.000 and 0.000). At these doses the perturbation controls the semantic answer and the measured truth rate follows from the true bit. This high-dose effect explains the tie between directions without identifying their unperturbed causal semantics.

Both directions on the inferred-bit task. In absolute terms they become nearly inert. The largest true-bit-1 range we can produce anywhere on the sweep is 0.077 (ally-fit) and 0.021 (mixed-fit), against 0.768 on the stated-bit task at the same layer. The more informative difference is not the magnitude but whether the response is *ordered* in α at all: the ally-fit direction still is, rising monotonically across the sweep (108 $\rightarrow$ 148 truthful trials of 517 from $\alpha_{\text{rel}} = -0.69$ to $+1.40$), whereas the mixed-fit direction falls weakly with α (134 at the negative end down to 123, before rebounding to 129 at the largest dose) and never leaves a 0.021 band. On this task the mixed-fit response is small and not ordered in α , so we do not report a potency ratio. The contrast does not isolate why the intervention has less influence on the inferred-bit task. The baseline differs (0.240 against 0.419), and the historical arm provenance is unresolved: the two saved Gemma inferred-bit sweeps record the steered model as emergent, while the fitted directions they load, and the run notes, record instructed. The original adapter identity was not retained. We therefore report the recorded sweep values without treating this as a controlled task-only comparison.

How the intervention relates to the diagnosis. Truth and prescribed action coincide on ally fitting data. That label equality alone does not determine which target the fitted direction follows outside that support. Here positive and negative unsigned perturbations move the two truth-conditioned correctness rates in opposition. This is compatible with multiple causal mechanisms, including a used truth-bit feature; it does not resolve the semantic ambiguity. The mixed-fit direction is a better truth readout in the tested contexts but generally has less effect on the answer at the compared sites and doses.

One thing we can *not* claim: the gap does not visibly grade with how often the policy lies. We see the same ordering at rival deception 1.000 as on the inferred-bit Qwen-14B arm at 0.639 and on the inferred-bit Gemma-9B sweep (whose steered arm is unresolved; 0.545 if it is the instructed arm), and the geometric analogue, the deep cosine of §K.1, is likewise flat across that range. Aliasing on the *fitting* data is a property of the protocol rather than of the policy, so there is no reason to have expected a gradient in the deception rate; we report the absence of one so that it is not read into the

results later. Identification and causal potency come apart: the better measurement of truth is the weaker lever on behaviour in these experiments. This does not diagnose the directions steered in other studies.

K.3 Depth dependence of the ally-fit advantage

The comparison above is at one layer, which leaves open whether the ally-fit advantage has anything to do with the two directions *differing*. §K.1 says they are nearly the same vector shallow and separate at depth, so that account makes a prediction: the advantage should be absent where $\cos \approx 1$ and present where the fits diverge. We ran the identical contrast at five layers, sweeping α over a $128 \times$ range at each because the median residual norm grows more than tenfold across the stack (83 at layer 8 to 860 at layer 40) and a grid fixed in raw α would not be comparable. The five dose–response pairs are plotted in Fig. 15.

layer	8	16	24	32	40
$\cos(\hat{d}_{\text{ally}}, \hat{d}_{\text{mixed}})$	0.996	0.986	0.773	0.603	0.540
ally-fit advantage	none	none	clear	clear	<i>both inert</i>

Where the two fits are nearly one vector. At layers 8 and 16, $\cos \geq 0.986$ and the two dose–response curves lie on top of each other (Fig. 15, first two panels), with no advantage either way, so the two have indistinguishable effects in this sweep there. Where they separate, at layers 24 and 32, it is the ally-fit direction that carries the behaviour: at layer 32 it drives the true-bit-1 rate from 0.16 to 0.93 across the sweep while the mixed-fit direction moves only 0.36 to 0.52. In these four layers, the ally-fit advantage appears where the fits have separated; layer 40 below shows that separation alone is insufficient.

We report this as an ordering and not as a potency ratio. A ratio here is not a stable quantity: at small doses the mixed-fit displacement is a noise-level denominator, at the largest doses at layer 24 both arms saturate, which sends any ratio to 1 by construction, and at that layer there are also doses where the denominator points the *opposite* way so the ratio is undefined rather than large. Quoting a multiplier, or worse a multiplier read off one chosen dose, would put a number on the claim that the underlying data do not support, the same failure we flag for the inferred-bit cell in §K.2. The two curves per layer are shown instead, so a reader can see the denominator that a ratio would hide.

Layer 40, the informative exception. Its cosine is the *lowest* we measure (0.540), so a simple “more separation, more advantage” rule predicts the largest effect there. Instead *both* directions are inert: the mixed-fit spread is slightly negative at every dose and the ally-fit spread never exceeds 0.036. Layer 40 of 42 has less downstream computation than the earlier sites, but this does not identify the cause of the small effect. In this measured sweep, fitted-direction separation alone is insufficient to predict a strong intervention response. Reading the effect off a single chosen dose would make the five layers look monotone in the cosine, and they are not.

Scope of the depth comparison. The ally-fit advantage accompanies direction separation at layers 24 and 32, but the layer-40 exception rules out a monotone relation with cosine across depth. Across the compared arms at fixed depth, neither the deep cosine nor the advantage visibly grades with deception rate. These observations do not establish a general predictor of intervention potency.

Ablation on the ally-fit direction is void exactly as before (post-ablation decodability 0.968 against the chosen 0.600 tolerance), so nothing in this section rests on an ablation. The Qwen arm sits on a 0.000 floor, so its negative half is uninformative by construction. And these are genuinely different axes rather than one axis re-signed (§K.1).

The ablation result is not a null; it is uninterpretable. Projecting out the direction leaves behaviour unchanged, but our positive control, specified before the run, shows the intervention did not remove the information: a freshly fitted probe still recovers the bit at 1.000 from the ablated activations, against our chosen tolerance of 0.600. Decodability here is $\max(A, 1 - A)$ for a freshly fitted probe’s AUROC A ; each label-permutation null is the 95th percentile of that quantity over 20 label permutations, each refitted on a stratified 70/30 split, and at layer 32 the nulls are 0.576–0.581. The same holds for every rank of a subspace ablation by iterative nullspace projection (INLP; Ravfogel et al., 2020) we could complete under an *underdetermined* fit, up to $k=8$ (Fig. 14). Re-running the fit *determined* ($n=4000 > d=3584$) changes this at $k=2$: post-ablation decodability falls to 0.600

against 0.727 for the underdetermined fit at the same rank, so the underdetermined subspace was under-removing by 0.128, exactly the bias its own guard predicts, and the reason we do not report the underdetermined curve as a bound. We completed the determined ladder at both layers, and the result is that no rank at either layer licenses a causal null. At layer 32, post-ablation decodability is 1.000, 0.5996, 0.5911 and 0.6077 at $k = 1, 2, 4, 8$; at layer 24 it is 1.000, 0.939, 0.913 and 0.997. Behaviour is unchanged throughout (ally truth rate 1.000).

Two of the eight cells clear the reported ceiling. That ceiling is $\max(\text{a tolerance we chose, the label-permutation null})$, i.e. the *more permissive* of the two, and our tolerance (0.600) is the looser one at every cell. Measured against its own permutation null instead, every cell at both layers sits above it: at layer 32 the nulls are 0.576, 0.580, 0.576 and 0.581, so even $k=2$ and $k=4$, which clear 0.600 by 0.0004 and 0.009, are 0.020 and 0.015 *above* the calibrated band, and $k=8$ fails both criteria outright. So the pass/fail language is an artifact of the permissive max, and we drop it. The bit remains partly decodable after every ablation we ran, so a flat behavioural rate is never licensed here as evidence about causal use.

What the completed ladder buys. A quantitative depth contrast, which is weaker than a pass/fail one. The layer-32 ablation drives decodability down to 0.59–0.61 while the layer-24 ablation leaves it at 0.91–1.00, so this removal procedure leaves far more of the bit decodable at layer 24, but neither layer yields an interpretable null, and the two layer-32 cells that met the tolerance are *bounded by void cells on both sides* ($k=1$ and $k=8$), so they are a middle band rather than a trend. Removal does not keep improving with rank: $k=8$ is worse than $k=4$ at layer 32 (0.608 vs 0.591) and worse at layer 24 as well (0.997 vs 0.913). One caveat travels with the whole ladder: the positive control’s permutation null sits near 0.58 rather than 0.50. A 95th percentile of $\max(A, 1 - A)$ exceeds 0.5 under finite-sample noise alone, and refitting at $n=600$ against a 3584-dimensional residual may raise it further; we have not separated the two. The calibrated null absorbs both, which is exactly why it, and not our tolerance, is the criterion we now report against. A behavioural null is only evidence about causal use if the ablation worked, so we draw no causal conclusion from it. What it does show is a fact about dimensionality: the bit is not confined to that rank-1 subspace, since removing the direction leaves it fully decodable from the orthogonal complement.

So the honest position is that the bit is linearly present and the mixed-fit direction we propose has weaker or less consistently ordered influence in the tested conditions than the ally-fit direction (§K.2); the ablation arm is uninterpretable rather than negative. Four limits on reading even that. Whole-residual patching *does* flip answers at 1.000 (App. F), so information at this site is used by something, though that intervention replaces every role-, prompt-, action- and truth-associated component at once, so it says nothing about how many dimensions carry the effect. Our α range initially reached 8.4–8.6% of the median residual norm across these two arms. We extended it sixteenfold, to $1.35\text{--}1.37\times$ the median residual norm, with an in-run $\alpha=0$ control passing through the same hook on the same episodes. On the saturated arm the positive and oracle-signed mixed-fit sweeps leave the rival-truth rate at 0.000 through $1.35\times$ the median residual norm; the negative unsigned sweep is also 0.000 except at its largest dose, where 2 of 1000 answers become truthful. The null is therefore not explained by restricting this direction to small doses within the tested range; the ally-fit direction does move answers at comparable doses (Fig. 14).

On the instructed arm the extended sweep does produce movement. The rival-truth rate rises from a measured baseline of 0.211 to 0.264 at $\alpha_{\text{rel}} = 0.69$ (nominally significant), then *collapses back* to 0.217 at 1.37. Before the extended sweep we fixed three criteria for counting a move as steering (monotone in α , sign-appropriate under an oracle sign, and not mirrored by the unsigned arm), and it fails all three: the curve is non-monotone, and the signed and unsigned arms are identical to four decimals (which, as noted above, the construction forces on exactly the trials where the movement occurs). The class-conditioned breakdown shows that on true-bit-0 trials the rate is exactly 0.000 at every α in both directions, while all observed changes occur on true-bit-1 trials. The positive and negative unsigned sweep in §K.2 confirms that $-\alpha d$ lowers the true-bit-1 rate. This establishes a signed response to the perturbation, not whether the underlying feature is a truth representation or an answer mechanism. Randomized codebooks also prevent interpreting semantic answer bias as a preference for one fixed character. This is one site in one layer of one model family. Finally, App. F’s “no arm difference” is stated at the strength each measure supports, which is not uniform. On the answer-position readout the matched difference is an *exact* tie: every one of the 8 matched cells reads 0.000 in both arms, so the paired difference has zero variance and a conventional equivalence test is undefined rather than passed; what bounds it is saturation, not statistics. On the two matched

measures that are *not* saturated, a two-one-sided-tests procedure does establish equivalence at a ± 0.05 tolerance (readout direction cosine, mean difference $+0.021$, $p = 0.034$; rival entropy, mean difference -0.003 , $p < 10^{-6}$). On the off-position readout it does **not**, for the reason given in App. F. We prefer reporting this unevenly to reporting a single tolerance that would be true of some measures and false of others.

L Supplementary figures

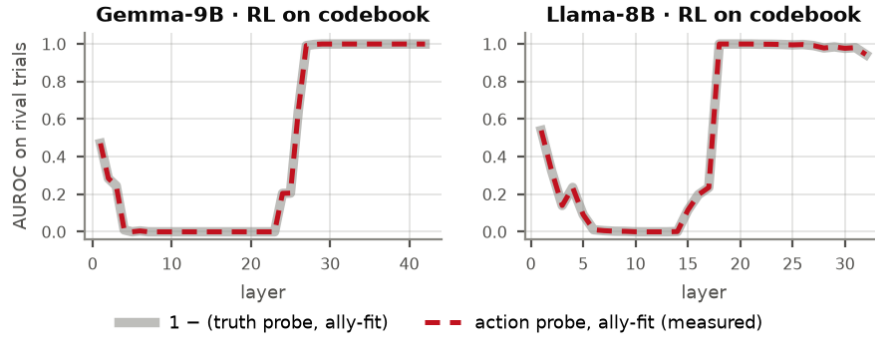
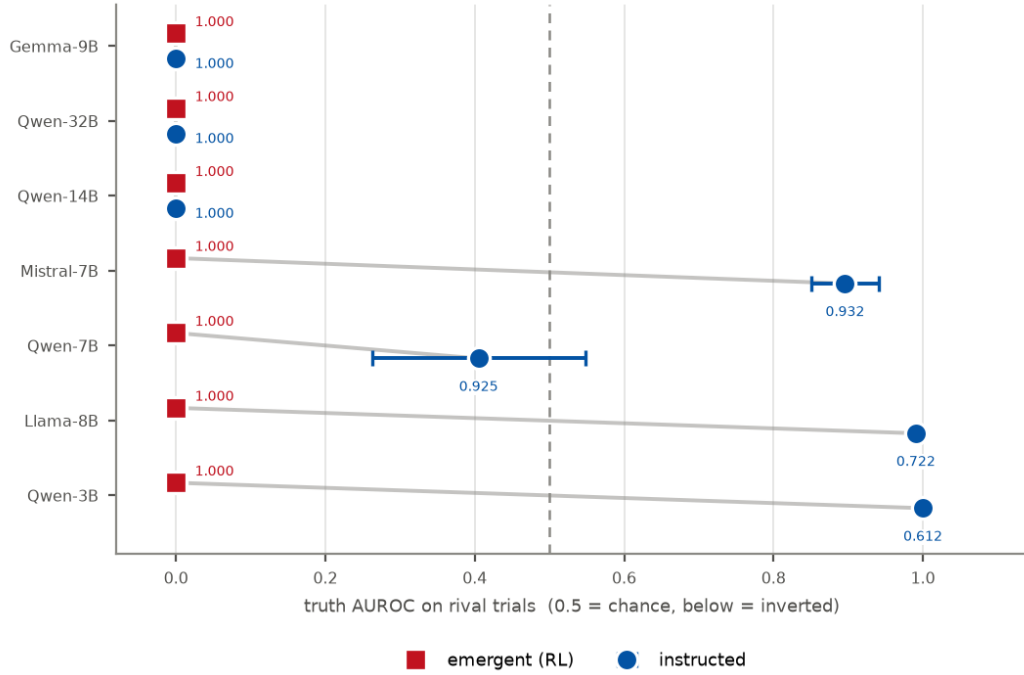


Figure 6: **The identity is forced.** The measured ally-fit *action* probe (dashed) lies exactly on $1 -$ (ally-fit *truth* probe) (thick grey) at every layer, because on ally data the two label vectors are identical and the two “probes” are one fit scored against opposite labels. Maximum deviation over all 751 (cell, layer) pairs: 2.2×10^{-16} .



numbers beside each marker = that arm's measured rival lie rate
 markers = mean over 3 seeds, error bars ± 1 SD (max 0.143)

Figure 7: **Inversion occurs under both training routes in the behaviourally matched comparisons.** The behaviourally-matched null of App. F. One row per model family; within a row the **red square** is the reward-trained (emergent) arm and the **blue circle** the instructed arm, dodged vertically so families where both sit at exactly the same AUROC still show two markers. Grey segments join the two arms of each family; their horizontal extent is the AUROC gap, while the vertical offset only separates the markers. Horizontal position is final-layer truth AUROC on held-out rival trials, so the dashed line at 0.5 is chance and everything left of it is inverted. Markers are the mean over 3 seeds with ± 1 SD error bars, and the number beside each marker is that arm's *measured rival lie rate*, not its AUROC. Rows are ordered by the instructed arm's lie rate, the variable a rate-based account would make the operative one. Read it as follows: every emergent arm lies on 1.000 of rival trials and every one reads 0.000; the three instructed arms that also lie on 1.000 (Gemma-9B, Qwen-32B, Qwen-14B) read 0.000 *too*, while the instructed arms that lie less reliably have variable readouts, including Qwen-7B seeds on both sides of chance. The matched saturated arms therefore refute an emergent-specific interpretation of inversion. Rival-deception saturation alone neither characterizes every inverted seed nor predicts inversion: the arm at rival deception 1.000 that does not invert (App. G) answers allies correctly on only 0.530 of trials, and we found no final-layer counterexample among cells meeting both reliability thresholds of §3. This is a failure to reproduce the route difference rather than demonstrated equivalence. The error bars are 0 in every saturated cell because those readouts sit at an extreme of the empirical ranking statistic; zero observed spread is not a tight population bound.

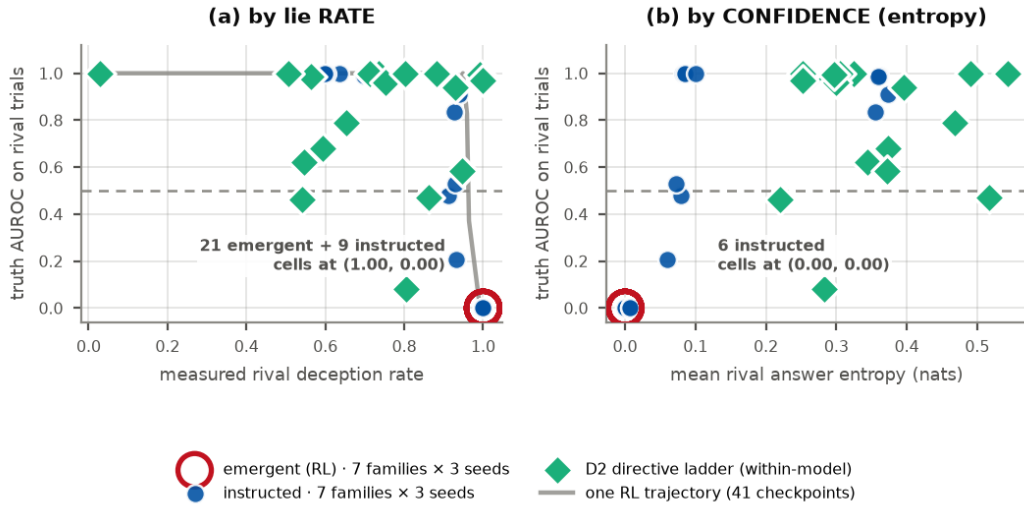


Figure 8: Neither lie rate (a) nor answer confidence, measured as mean rival-answer entropy (b), orders the readout: cells with the same behaviour sit at both 0.00 and 1.00. Both were hypotheses of ours, and both are refuted (App. G). Red rings and blue circles show the reward-trained and instructed grid cells; green diamonds show the within-model directive variants. The grey line joins one RL run’s 41 checkpoints in horizontal-axis order, not time order. Dashed horizontal lines mark chance.

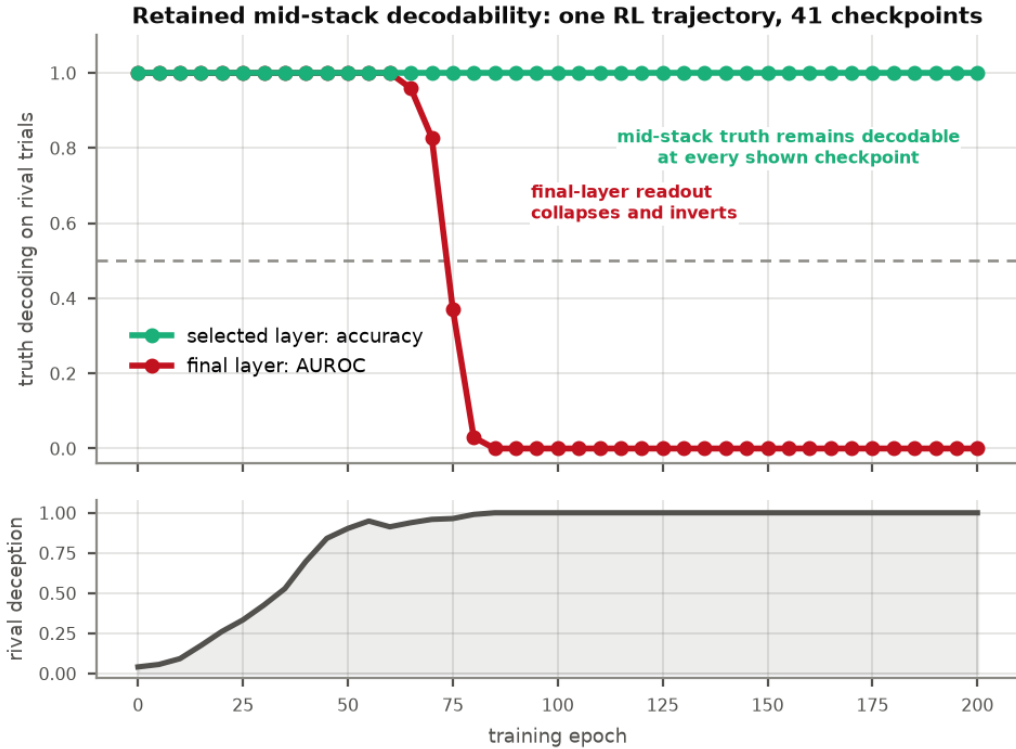


Figure 9: Retained mid-stack decodability: one RL trajectory, 41 checkpoints. Top: rival truth-classification accuracy at the layer selected by highest ally-validation accuracy (green), and rival truth AUROC at the final layer, at the answer-prediction position (red). These are different metrics; the selected-layer accuracy stays pinned while the final-layer ranking collapses and inverts. Bottom: measured rival deception rate over the same training epochs.

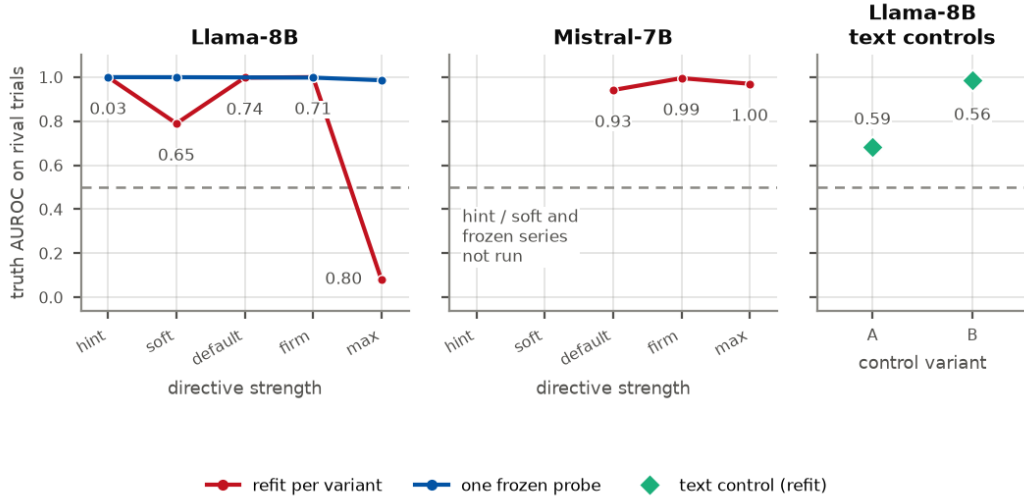


Figure 10: **The directive-strength ladder moves the refit probe, not the frozen one.** Left and centre: truth AUROC on rival trials as the rival directive strengthens from *hint* to *max*, with model weights held fixed. Red: a probe fit separately on each variant’s ally data. Blue: one probe fit on the reference variant and cross-scored, available only for Llama-8B. Lines connect the measured rungs; Mistral’s two weakest rungs were not run. Right: two separate Llama-8B text controls, scored with refit probes. A removes negation words from the max directive; B pads the default rule to approximately the max directive’s length. These controls are not additional strength rungs. Numbers beside markers are measured rival deception rates, not AUROCs: 0.59 and 0.56 for A and B versus 0.80 for max, so the controls are not behaviour matched. Dashed horizontal lines mark chance. $N=1000$ collected episodes per variant; see §5 and App. G.

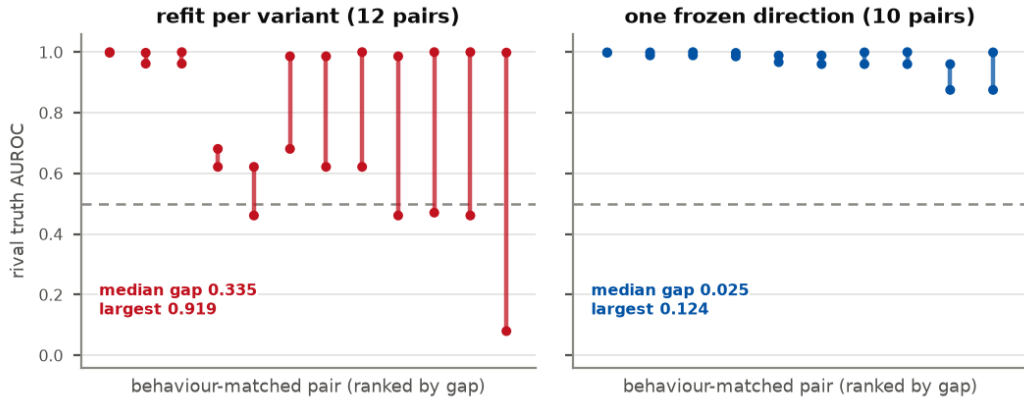


Figure 11: **Behaviour-matched prompt pairs: refitting splits them, one frozen direction does not.** Every pair of the 15 ladder rungs (the reference plus 14 variants) whose rival deception rates agree to within 0.05, each drawn as a segment joining the two members’ rival truth AUROCs, ranked by gap. Left: a probe refit on each variant’s own ally data, median gap 0.335, largest 0.919, and five pairs straddle chance. Right: one direction fit once on the reference variant and cross-scored, median gap 0.025, largest 0.124. Two pairs contain the reference variant itself, which has no frozen score by construction, so the right panel has 10 pairs to the left panel’s 12. Pairs are enumerated under a stated tolerance, not selected on the size of their gap.

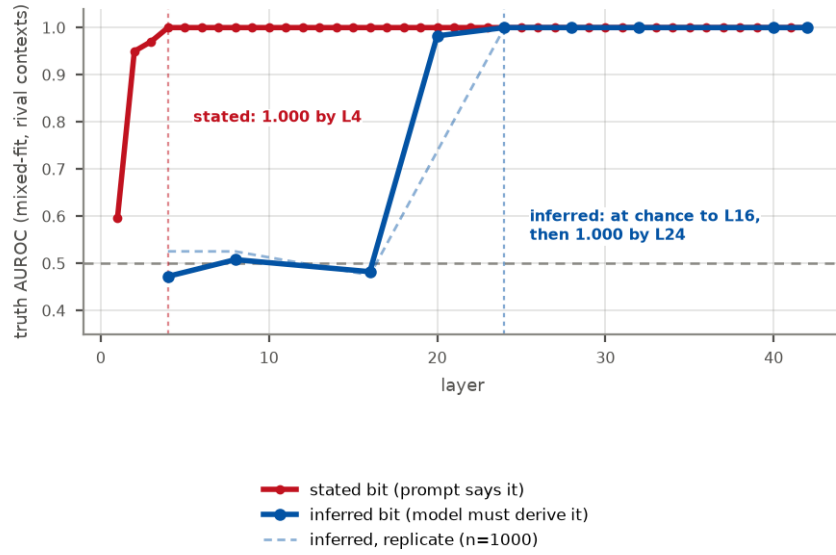


Figure 12: **The depth step.** Truth AUROC of the mixed-fit probe on held-out rival trials, Gemma-9B. When the prompt *states* the bit (red) it is linearly readable almost immediately and reaches 1.000 by layer 4, consistent with reading a fact stated in the prompt. When the model must *derive* the bit (blue, $N=2000$; faint dashed is the $N=1000$ replicate) the same protocol sits at chance through layer 16 and only then rises, reaching 1.000 by layer 24. The contrast is evidence against reading the deep readout as a direct copy of a prompt token; it does not identify how the bit is computed (§8). The inferred-bit curves use sparse layer grids with gaps of up to eight layers; transitions inside those gaps are not resolved.

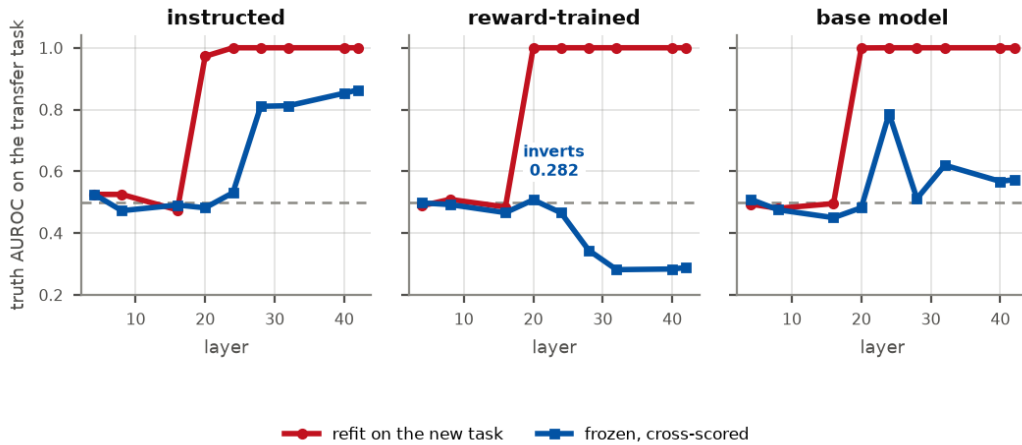


Figure 13: **A refitted probe does not measure transfer.** Each panel scores the inferred-bit task. Red: a probe *refit* on that task; it reaches 1.000 by layer 24 in all three arms (instructed, the stated-bit reward-trained adapter and the base model; 0.97–1.000 at layer 20). Blue: a mixed-fit direction fitted on the stated-bit task and merely cross-scored (the reward-trained arm’s own in its panel, the instructed arm’s in the other two). It never exceeds 0.864 on the instructed arm, and on the reward-trained arm, which deceives on only 0.235 of rival trials of this task, it falls to a minimum of 0.282 at layer 32. A refit number and a frozen number answer different questions, and only the frozen one bears on transfer.

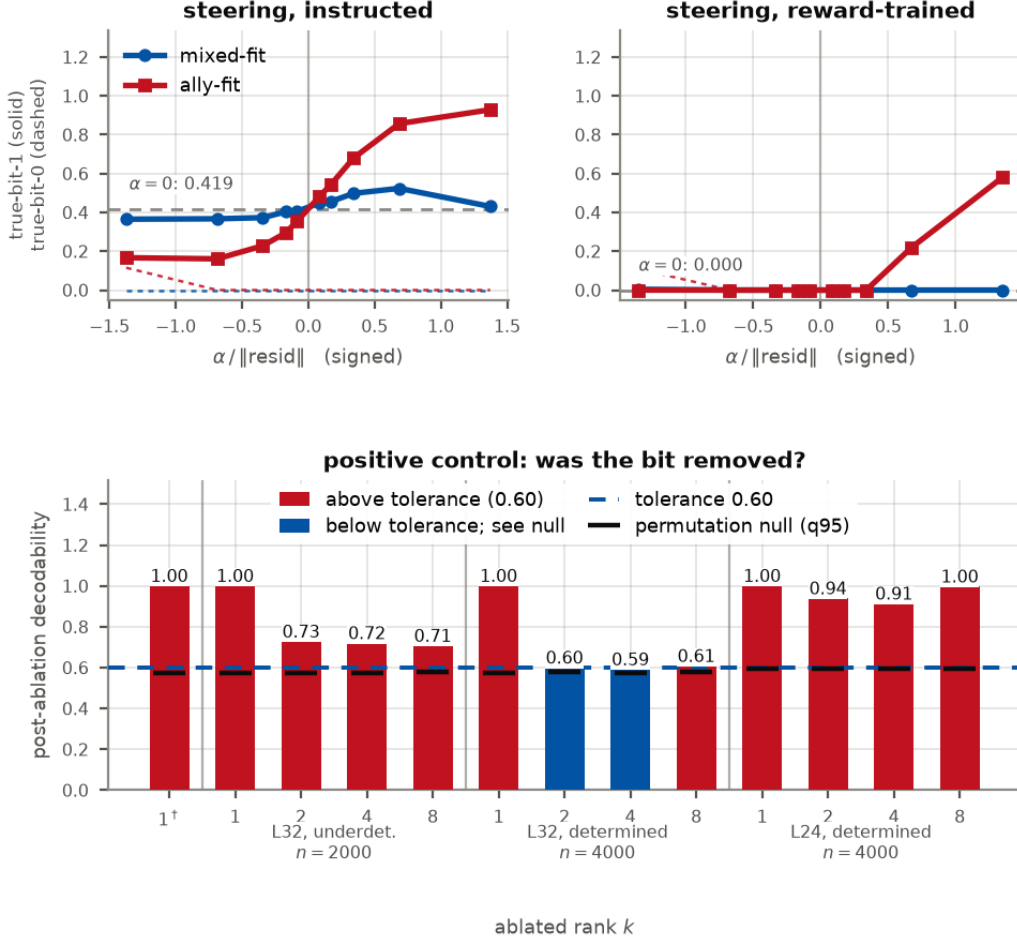


Figure 14: **At the tested layers and doses, the ally-fit direction has the larger behavioural effect.** Top left and right: steering at layer 32, swept *through zero* against a single in-run $\alpha=0$ baseline. Solid lines are the true-bit-1 rate, dashed the true-bit-0 rate. The mixed-fit direction (blue) responds weakly, $0.365 \rightarrow 0.524$ before degrading at the largest dose; the ally-fit direction (red, the one we criticize) swings from 0.167 at the most negative dose to 0.929 (minimum 0.161), and on the saturated arm breaks a floor the mixed-fit direction moves by at most 2 of 1000 answers. True-bit-0 stays pinned at 0.000 almost everywhere, lifting only under the strongest negative ally-fit push. These unsigned, class-conditioned effects do not distinguish a causally used truth-bit feature from another mechanism producing semantic answer bias. Bottom: the positive control, specified before the run. The 1[†] bar is the separate mixed-fit rank-one ablation run at layer 32; the subsequent rank-one bars are the first steps of the iterative subspace-ablation ladders. Post-ablation decodability must fall *below* the ceiling (0.600) for a behavioural null to be interpretable; red bars fail and are void. Both determined ladders are complete, and no cell at either layer falls below its own permutation null (black dashes). Ranks 2 and 4 at layer 32 clear the 0.600 tolerance by 0.0004 and 0.009, but sit 0.020 and 0.015 *above* the calibrated null, and $k=8$ (0.608) fails both, so the two clearing cells are a middle band bounded by void cells at $k=1$ and $k=8$. The contrast with layer 24 is one of degree (0.59–0.61 versus 0.91–1.00), not of pass versus fail. Ranks 16 and 32 of the underdetermined layer-32 ladder did not complete and are not plotted.

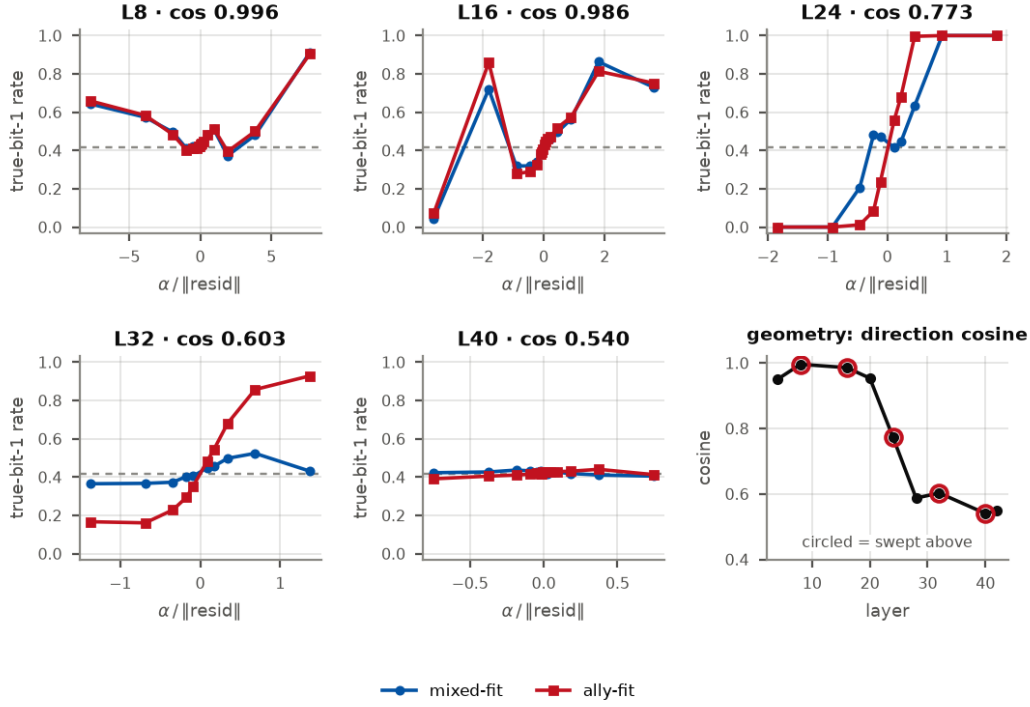


Figure 15: **The ally-fit advantage appears only where the two fitted directions differ, and separation alone is not enough.** Each panel is the same causal contrast run at one depth on the Gemma-2-9B instructed arm: true-bit-1 rate against dose, mixed-fit direction in blue, the criticized ally-fit direction in red, grey dashed line the in-run $\alpha=0$ baseline. At layers 8 and 16, where $\cos \geq 0.986$, the two curves lie on top of each other: the directions are nearly one vector and have indistinguishable effects in this sweep. At layers 24 and 32 they separate, and it is the ally-fit direction that carries the behaviour ($0.16 \rightarrow 0.93$ at layer 32 while the mixed-fit direction moves $0.36 \rightarrow 0.52$). At layer 24 both saturate to 1.000 at the largest doses, so the apparent tie there is saturation rather than equivalence. Layer 40 is the informative exception: it has the lowest cosine we measure yet both arms are flat. In this sweep separation alone does not ensure a large intervention effect; the remaining downstream computation does not establish why the effect is small. Bottom right: the geometry the five panels are keyed to, with the swept layers circled. *We plot both dose-response curves rather than their ratio*: at small doses the mixed-fit displacement is a noise-level denominator, so a ratio is unstable, and where both arms saturate at large doses (layer 24) it is 1 by construction. Note the horizontal scales differ by panel: the median residual norm grows from 83 at layer 8 to 860 at layer 40, so a fixed α grid spans a much wider $\alpha / \|\text{resid}\|$ range shallow, and the outermost shallow doses are representation damage rather than dose-response.